

RAINBOW NORMAL FORMS AND THE L-SPACE PROPERTY OF ONE-RAINBOW $(1, 1)$ -KNOTS

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ABSTRACT. A $(1, 1)$ -knot admits a doubly pointed genus-one diagram $(T^2, \alpha, \beta, z, w)$ [8, 12]; cutting along α presents β as $p = |\alpha \cap \beta|$ arcs in an annulus, and Doyle's rainbow normal form $[p, a, b, r]$ records a rainbows per wall, $p - 2a$ bands of one step size b , and an offset r [5, 6]. We treat $a = 1$, where S^3 -admissibility is the pair of conditions $|e| = 1$ on the drift $e = \beta \cdot \alpha$ and $|\tau^{\mathbb{Z}}(L_1)| = p$ on the transition word.

Every admissible one-rainbow diagram is reduced and coherent in the sense of [7], so each bigon of $T^2 \setminus (\alpha \cup \beta)$ covers z or w ; since $\text{Sym}^1(T^2) = T^2$, every index-one class is an embedded bigon with a single flow line [9, 12], the differential is the adjacency matrix of the empty-bigon relation over $\mathbb{F} = \mathbb{Z}/2$, and

$$\widehat{\partial} = \mathbf{0}, \quad \text{rk}(\widehat{HFK}(K)) = p, \quad \Delta_K(t) = \sum_{j=1}^p (-1)^{j+1} t^{A_j}, \quad A_1 > \dots > A_p,$$

with x_z topmost for $e > 0$ and x_w for $e < 0$, so K is an L-space knot [10].

The census satisfies $|\mathcal{C}_p| = 4, 6, 8, 8, 10, 12, 14, 16$ for $p = 3, 5, \dots, 17$ and contains $T(2, p)$ and $T(\frac{p+1}{2}, \frac{p+3}{2})$; for 40 of the 48 tuples with $p \leq 13$ the semigroup $S_K = \langle a, b \rangle$ is generated by two elements, $|\mathcal{G}_K| = g = \frac{(a-1)(b-1)}{2}$ and $K = T(a, b)$ [15, 13], while the other eight are L-space knots that are not algebraic [1, 2]. In the algebraic case $\mathcal{G}_K = d(\cdot; a, b)^{-1}(0)$ and

$$\nabla_a \nabla_b d = \delta_0, \quad \nabla_a \nabla_b \frac{u_{\pm}}{ab} = B_{(a,b)}, \quad t^g \Delta_K(t) = (1-t)(1-t^{ab}) \sum_{n \geq 0} d(n; a, b) t^n,$$

so the Alexander polynomial is a second difference of a truncated power, separated from the box spline $B_{(a,b)}$ by an ab -periodic error of modulus at most one [11, 4, 3].

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1. INTRODUCTION

1.1. Rainbow normal forms. A knot $K \subset S^3$ is a $(1, 1)$ -knot when it admits a doubly pointed genus-one diagram $(T^2, \alpha, \beta, z, w)$ [8, 12], and since $\text{Sym}^1(T^2) = T^2$ the generators of $\widehat{CFK}(K)$ are the $p = |\alpha \cap \beta|$ intersection points [9]. Every reduced such diagram is isotopic to one with α a meridian [12, 7]. Cutting along α splits β into *rainbow arcs* $\mathbb{D} \mid \mathbb{I} \mid \mathbb{Q}$, each cutting off a bigon that carries a basepoint, and *vertical arcs* $\downarrow \uparrow$. Doyle's normal form $[p, a, b, r]$ [5, 6] records a rainbow arcs per wall, b vertical arcs of one step size, and an offset r ; throughout $a = 1$, so $\mathbb{D} \mid \mathbb{I} \mid \mathbb{Q}$ carries a single rainbow arc on each wall.

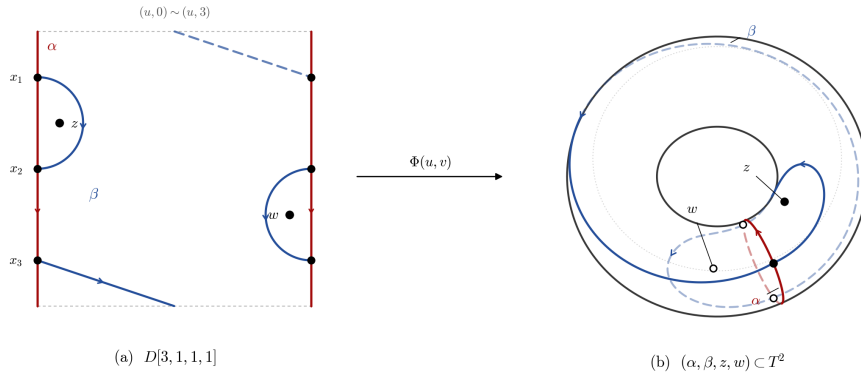


FIGURE 1. $[3, 1, 1, 1]$, cut open along α and reglued by Φ : one rainbow arc per wall carrying z and w , one vertical arc between them, $e = +1$, $K = T(2, 3)$. Behind T^2 dashed.

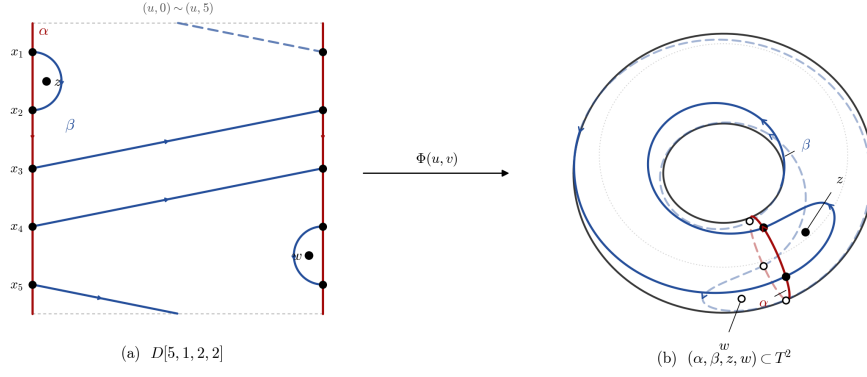


FIGURE 2. $[5, 1, 2, 2]$ as in Figure 1, with three vertical arcs and β connected; $e = -1$, $K = T(3, 4)$.

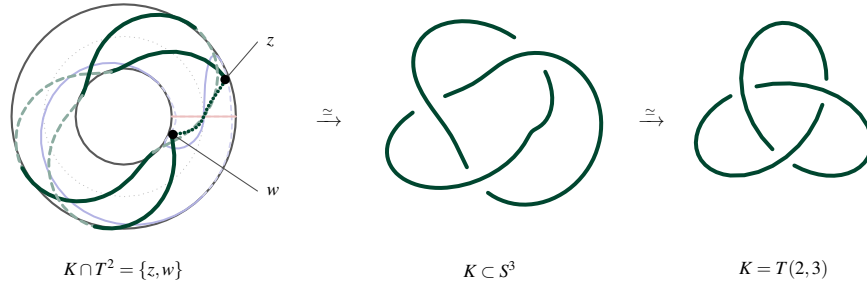


FIGURE 3. The knot of Figure 1 meets T^2 only at z and w , one arc on the surface and one dotted through the solid torus. Windings 2 and 3 give $K = T(2, 3)$.

1.2. Result. Ozsváth and Szabó showed that an L-space knot has $\widehat{HFK}$ of rank one in each Alexander grading it occupies, with alternating unit coefficients in Δ_K [10]. We call this the *Alexander criterion*. At $a = 1$ it holds for every admissible tuple, and the candidacy is realised.

Theorem A. Let $[p, 1, b, r]$ be S^3 -admissible in the sense of Definition 2.2.3 and present $K \subset S^3$. Then

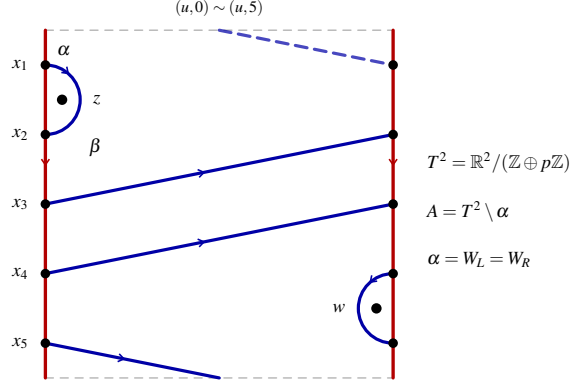
- (1) $\text{rk}(\widehat{HFK}(K)) = p$, with $A_1 > \dots > A_p$ pairwise distinct;
- (2) $\Delta_K(t) = \sum_{j=1}^p (-1)^{j+1} t^{A_j}$, so K satisfies the Alexander criterion;
- (3) the generator of topmost Alexander grading is x_z for $e > 0$ and x_w for $e < 0$;
- (4) K is an L-space knot.

Proof. §4. □

2. PRELIMINARIES

2.1. Rainbow normal form.

Definition 2.1.1 (Doubly pointed diagram). $(T^2, \alpha, \beta, z, w)$ with T^2 genus one, $\alpha, \beta \subset T^2$ essential simple closed curves meeting transversely, and $z, w \in T^2 \setminus (\alpha \cup \beta)$.



Definition 2.1.2 (Rainbow tuple, [5]). $[p, a, b, r] \in \mathbb{Z}^4$ with

$$p \geq 3 \text{ odd}, \quad 1 \leq a \leq \frac{p-1}{2}, \quad 0 \leq b \leq p-2a, \quad 1 \leq r \leq p,$$

so p generators, a rainbows per wall, b bands, first landing r .

Definition 2.1.3 (Cut annulus). $A = T^2 \setminus \alpha = [0, 1] \times \mathbb{R}/p\mathbb{Z}$, height increasing downwards,

$$W_L = \{0\} \times \mathbb{R}/p\mathbb{Z}, \quad W_R = \{1\} \times \mathbb{R}/p\mathbb{Z}, \quad L_h = \left(0, h - \frac{1}{2}\right), \quad R_h = \left(1, h - \frac{1}{2}\right), \quad h \in \mathbb{Z}/p\mathbb{Z},$$

with $\downarrow \downarrow: R_h \mapsto L_h$, both descending to $x_h \in \alpha \cap \beta$.

Definition 2.1.4 (Rainbow normal form, [5]). $D[p, a, b, r] = \mathbb{D} \mathbb{Q}$ is Definition 2.1.3 with $\beta \cap A$ the p disjoint embedded arcs

$$\begin{aligned} L_i &\leftrightarrow L_{2a+1-i}, & R_{r+b-1+i} &\leftrightarrow R_{r+b+2a-i}, & 1 \leq i \leq a, \\ L_s &\leftrightarrow R_{r+s-2a-1}, & & & 2a < s \leq 2a+b, \\ L_s &\leftrightarrow R_{r+s-1}, & & & 2a+b < s \leq p, \end{aligned}$$

the rainbows, the bands, and the returns, the last crossing $(u, 0) \sim (u, p)$ iff $r + s - 1 > p$, and

$$z \in L_a L_{a+1}, \quad w \in R_{r+b+a-1} R_{r+b+a},$$

the innermost rainbow bigons. The right endpoints $\rho: \{2a+1, \dots, p\} \rightarrow \mathbb{Z}/p\mathbb{Z} \setminus \{r+b, \dots, r+b+2a-1\}$ are a bijection.

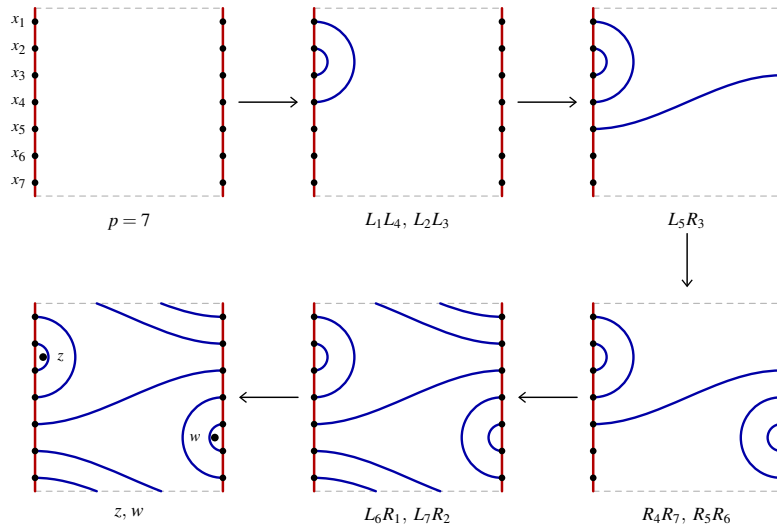
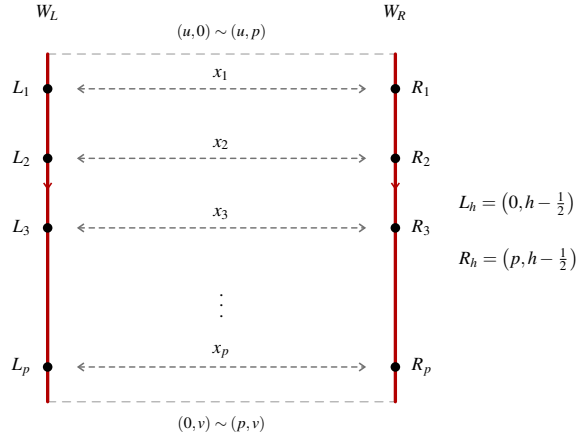


FIGURE 4. Fairchild's construction of $D[7, 2, 1, 3]$, one arc family at a time from the marked points to the basepoints. Drawn after [6].

Convention 2.1.5 (One rainbow). $a = 1$, $v = r + b$, so Definition 2.1.4 reads

$$L_1 \leftrightarrow L_2, \quad R_v \leftrightarrow R_{v+1}, \quad L_s \leftrightarrow R_{\rho(s)}, \quad \rho(s) = \begin{cases} r+s-3, & 3 \leq s \leq b+2, \\ r+s-1, & b+2 < s \leq p, \end{cases}$$

with $z \in L_1 L_2$, $w \in R_v R_{v+1}$, hence $x_z \in \{x_1, x_2\}$ and $x_w \in \{x_v, x_{v+1}\}$.



Definition 2.1.6 (Rainbow and vertical arcs, [12, 7]).

$$\mathbb{D} \mid = L_1 L_2, \quad \mid \mathbb{D} = R_v R_{v+1}$$

are the *rainbow arcs*, $\downarrow \uparrow = c_s = L_s R_{\rho(s)}$ the *vertical arcs*, $\mathbb{D}$ and $\mathbb{D}$ the two rainbow families, and $\mathbb{D}$ the b arcs with $\rho(s) = r + s - 3$. The innermost rainbow arcs are the *caps* $\mathbb{D}$, $\mathbb{D}$, cutting off

$$\mathbb{D}_z = D_z \subset W_L, \quad \mathbb{D}_w = D_w \subset W_R.$$

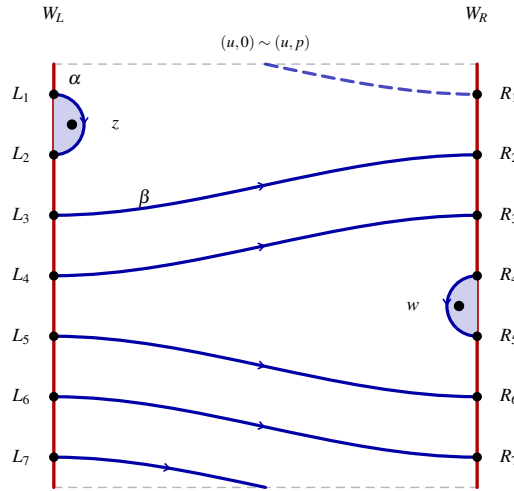


FIGURE 5. $\mathbb{D} \mathbb{D} = D[7, 1, 2, 2]$ with $v = 4$. The rainbow arcs $\mathbb{D} \mid = L_1 L_2$, $\mid \mathbb{D} = R_4 R_5$ cut off $\mathbb{D}_z$, $\mathbb{D}_w$; the bands $\mathbb{D}$ are $L_3 R_2$, $L_4 R_3$; the wrapping arcs are $L_5 R_6$, $L_6 R_7$; and $L_7 R_1$ crosses $(u, 0) \sim (u, p)$, its second piece dashed. Under $\downarrow \uparrow$ the slots L_h , R_h have common image x_h .

Definition 2.1.7 (Cap corners). $x_z \in \partial D_z$ and $x_w \in \partial D_w$.

Remark 2.1.8 (Standard position, [12, 7, 2]). α is the meridian and β carries the rainbow arcs.

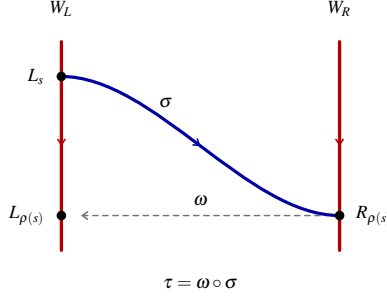
Proof. [12, §2]. □

2.2. Drift and admissibility.

Definition 2.2.1 (Traversal).

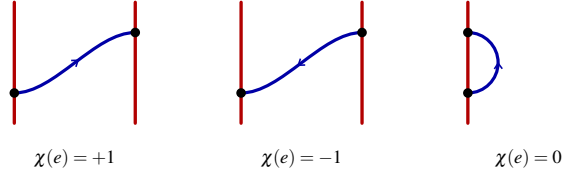
$$\omega(L_h) = R_h, \quad \omega(R_h) = L_h, \quad \tau = \omega \circ \sigma$$

for σ the matching of Definition 2.1.4. One step is $\downarrow \uparrow$, and the *traversal* is $\tau^{\mathbb{Z}}(L_1)$.

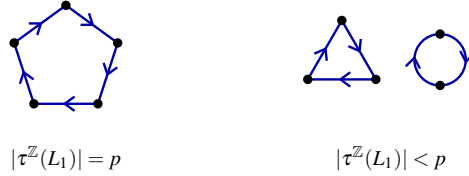


Definition 2.2.2 (Drift). For $e \in \beta \cap A$,

$$\chi(\downarrow \downarrow) = 0, \quad \chi(\downarrow \uparrow) = +1, \quad \chi(\uparrow \downarrow) = -1, \quad e = \sum_e \chi(e) = \beta \cdot \alpha.$$



Definition 2.2.3 (Admissibility). $D[p, 1, b, r]$ is S^3 -admissible when $\downarrow \uparrow$, that is $|\tau^{\mathbb{Z}}(L_1)| = p$ and $|e| = 1$.



Lemma 2.2.4 (Rank). $D[p, 1, b, r]$ admissible implies β connected, with $(T^2, \alpha, \beta) \simeq S^3$ and $\text{rk}(\widehat{CFK}(K)) = p$.

Proof.

$$\begin{aligned} |\tau^{\mathbb{Z}}(L_1)| = p &\xrightarrow{2.2.1} |\pi_0(\beta)| = 1 \\ \downarrow \downarrow \implies \alpha \cap \beta = \{x_1, \dots, x_p\} &\implies \text{rk}(\widehat{CFK}(K)) = p \\ |e| = 1 &\xleftrightarrow{2.2.2} |\beta \cdot \alpha| = 1 \xrightarrow{2.1.1} (T^2, \alpha, \beta) \simeq S^3 \end{aligned}$$

□

Remark 2.2.5 (Two step sizes). $\rho(s) - s \in \{r-3, r-1\}$, so τ is an exchange of two intervals $\downarrow \uparrow$, and $\star$ is a p -cycle exactly when $D[p, 1, b, r]$ is admissible.

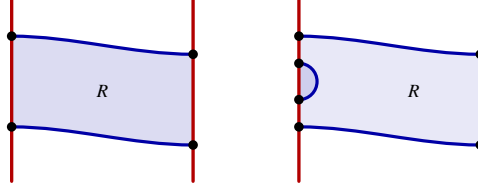
Proof. Definition 2.1.4 and Definition 2.2.3. □

2.3. Reduced and coherent diagrams.

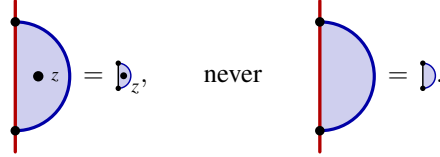
Definition 2.3.1 (Region type). A component R of $T^2 \setminus (\alpha \cup \beta)$ is n -sided when $|\partial R \cap \alpha \cap \beta| = 2n$,

$$\text{D} (n=1), \quad \text{D} (n=2), \quad \text{D} (n=3), \quad \text{D} (n=4),$$

with tints lightening as n grows. The darker lobe against a wall in D is a neighbouring bigon, one of the n sides of R and not part of it.



Definition 2.3.2 (Reduced diagram, [7]). A diagram is *reduced* $\mathbb{D}$ when every bigon of $T^2 \setminus (\alpha \cup \beta)$ carries a basepoint,

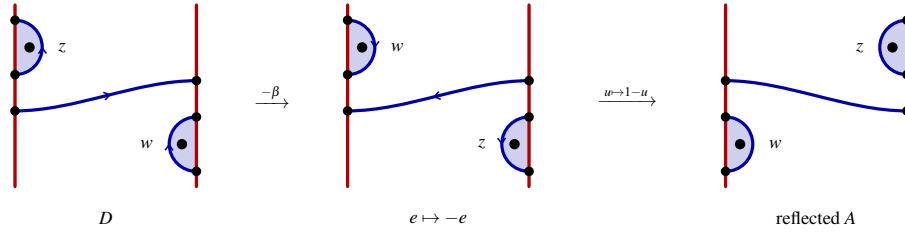


Definition 2.3.3 (Coherent diagram, [7]). A reduced diagram is *coherent* when α, β admit orientations coorienting every bigon, equivalently [2]



2.4. Symmetries.

Definition 2.4.1 (Orientation reversal). Reversing β sends $\downarrow \uparrow \mapsto \uparrow \downarrow$ and exchanges the basepoints $\mathbb{D} \leftrightarrow \mathbb{D}$; reflecting A is $\mathbb{D} \leftrightarrow \mathbb{D}$.



Lemma 2.4.2 (Effect on the gradings). Reversing β sends $e \mapsto -e$ and $A \mapsto -A$.

Proof.

$$\begin{aligned} \chi(\downarrow \uparrow) &= -\chi(\uparrow \downarrow) \xrightarrow{2.2.2} e \mapsto -e \\ \mathbb{D} \leftrightarrow \mathbb{D} &\implies (n_z, n_w) \mapsto (n_w, n_z) \implies A \mapsto -A \end{aligned}$$

□

2.5. Knot Floer input.

Theorem 2.5.1 (Coherence criterion, [7, Theorem 1.2]). A reduced (1,1)-diagram presents an L-space knot if and only if it is coherent.

Proof. [7, Theorem 1.2]

□

Theorem 2.5.2 (Alexander polynomial, [10, Theorem 1.2]). For $K \subset S^3$ an L-space knot,

$$\Delta_K(t) = \sum_{j=0}^{2n} (-1)^j t^{A_j}, \quad A_0 > A_1 > \cdots > A_{2n},$$

and $\widehat{\text{HFK}}(K)$ has rank one in each A_j and vanishes elsewhere.

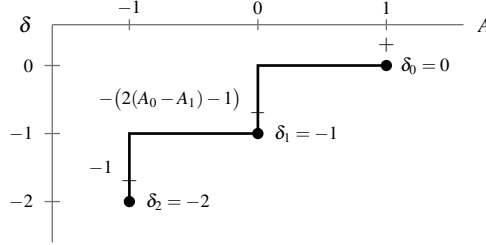
Proof. [10, Theorem 1.2]

□

Definition 2.5.3 (Alexander Criterion). Δ_K has the form of Theorem 2.5.2.

Definition 2.5.4 (Maslov normalisation). The top Alexander generator has $M = 0$, and

$$\delta_0 = 0, \quad \delta_j = \begin{cases} \delta_{j-1} - 2(A_{j-1} - A_j) + 1, & j \text{ odd}, \\ \delta_{j-1} - 1, & j \text{ even}. \end{cases}$$



$$K = T(2,3), \quad \Delta_K(t) = t - 1 + t^{-1}$$

Remark 2.5.5 (Mirrors). $\mathbb{D} \wr \mathbb{Q}$ reverses $\delta_0, \dots, \delta_{2n}$, so Definition 2.5.4 precedes Definition 2.5.3.

Proof. Lemma 2.4.2. □

2.6. Combinatorial input.

Definition 2.6.1 (Semigroup and staircase). For $\gcd(\{a, b\}) = 1$,

$$S = \langle a, b \rangle = \{ia + jb \mid i, j \geq 0\}, \quad \mathcal{G} = \mathbb{Z}_{\geq 0} \setminus S, \quad |\mathcal{G}| = g, \quad \max(\mathcal{G}) = ab - a - b = 2g - 1.$$

The *staircase* $\sqsubset$ steps East at each $s \in S$ and South at each $\circ$, for $0 \leq n \leq 2g - 1$.

Definition 2.6.2 (Vector partition function and box spline).

$$d(n; a, b) = |\{(i, j) \in \mathbb{Z}_{\geq 0}^2 \mid ia + jb = n\}|, \quad B_{(a,b)} = \sqsubset = (ax + by)_* (dx dy|_{[0,1]^2}).$$

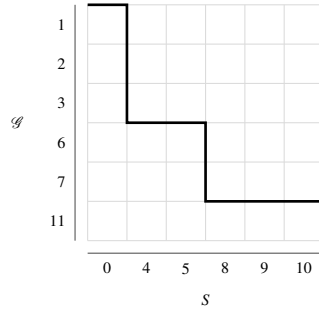


FIGURE 6. The staircase $\sqsubset$ of $S = \langle 4, 5 \rangle$, $g = 6$. Horizontal steps are read off the S -axis and vertical steps off the $\mathcal{G}$ -axis, so the axes list $S \cap [0, 2g) = \{0, 4, 5, 8, 9, 10\}$ and $\mathcal{G} = \{1, 2, 3, 6, 7, 11\}$.

Lemma 2.6.3 (Staircase and support). For $K = T(a, b)$,

$$\Delta_K(t) = t^{-g}(1-t) \sum_{s \in S, s < 2g} t^s, \quad \text{Supp}(\Delta_K) = \{A_0, \dots, A_{2n}\}.$$

Proof.

$$\Delta_{T(a,b)}(t) = \frac{(t^{ab} - 1)(t - 1)}{(t^a - 1)(t^b - 1)} \implies \Delta_{T(a,b)}(t) = (1 - t) \sum_{s \in S} t^s$$

$$\max(\mathcal{G}) = 2g - 1 \implies \sum_{s \in S} t^s \equiv \sum_{s \in S, s < 2g} t^s \pmod{t^{2g}} \xrightarrow{2.5.2} \text{Supp}(\Delta_K) = \{A_0, \dots, A_{2n}\}$$

□

3. REDUCEDNESS AND COHERENCE

3.1. **Reducedness.** Throughout $D = D[p, 1, b, r]$, and $\beta \cap A$ consists of the arcs of Definition 2.1.6,

$$\mathcal{D} = L_1 L_2, \quad \mathcal{C} = R_v R_{v+1}, \quad \mathcal{C}_s = L_s R_{p(s)}, \quad 3 \leq s \leq p.$$

Lemma 3.1.1 (Bigon classification). *The embedded bigons of $D[p, 1, b, r]$ are $\mathcal{D}_z$ and $\mathcal{C}_w$.*

Proof.

$$\begin{aligned} \partial \mathcal{D} &= a \cup b, \quad a \subset \alpha, \quad b \subset \beta \cap A \xrightarrow{2.1.4} b \in \{L_1 L_2, R_v R_{v+1}, c_s\} \\ b = c_s &\xrightarrow{7} [a \cup c_s] \neq 0 \in H_1(T^2) \implies a \cup c_s \neq \partial \mathcal{D} \xrightarrow{8} c_s \subset \partial R, \quad |\partial R \cap \alpha \cap \beta| \geq 4 \\ &\implies b \in \{L_1 L_2, R_v R_{v+1}\} \xrightarrow{2.1.6} \mathcal{D} \in \{\mathcal{D}_z, \mathcal{C}_w\} \end{aligned}$$

□

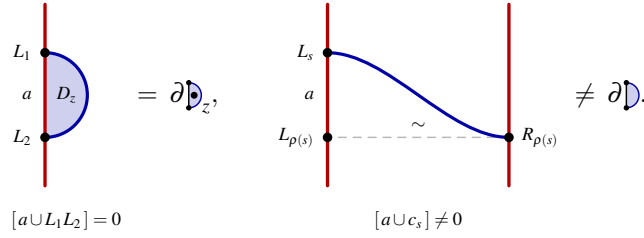


FIGURE 7. The α segment a closes a rainbow arc to a null class and a vertical arc to an essential one in $H_1(T^2)$.

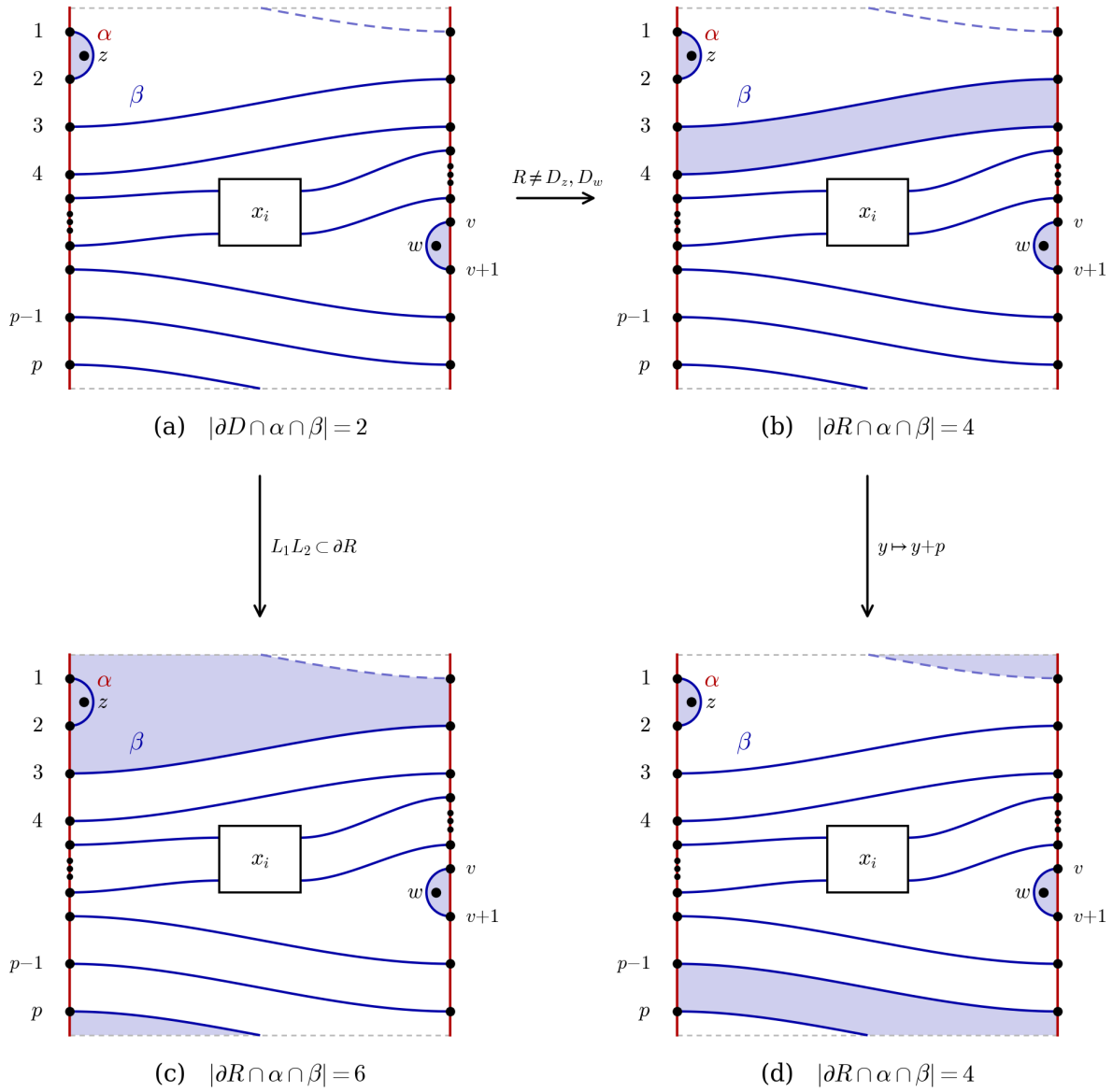
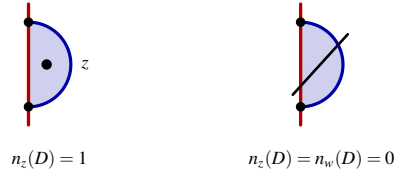


FIGURE 8. Corner counts, 2 for the disk in (a) and ≥ 4 for the regions in (b)–(d).

Reducedness asks that no bigon be empty, since an empty one is a Maslov index one domain with $n_z = n_w = 0$.



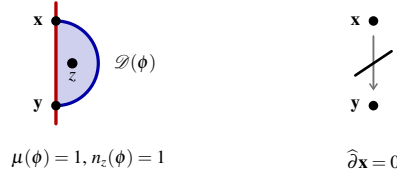
Corollary 3.1.2 (Reducedness). $D[p, 1, b, r]$ is reduced ∇ , every bigon carrying a basepoint.

Proof.

$$\mathbb{D} \text{ embedded} \xrightarrow{3.1.1} \mathbb{D} \in \{\mathbb{D}_z, \mathbb{D}_w\} \implies z \in \mathbb{D} \text{ or } w \in \mathbb{D} \xLeftrightarrow{2.3.2} D \text{ reduced}$$

□

Generators $\mathbf{x}, \mathbf{y} \in \alpha \cap \beta$ span $\widehat{CFK}(K)$, and $\widehat{\partial}$ is the matrix whose $(\mathbf{y}, \mathbf{x})$ entry counts the pseudoholomorphic Whitney disks from $\mathbf{x}$ to $\mathbf{y}$ modulo the $\mathbb{R}$ -action. Since $\text{Sym}^1(T^2) = T^2$, each generator is a single point. A class $\phi \in \pi_2(\mathbf{x}, \mathbf{y})$ has Maslov index $\mu(\phi)$, domain $\mathcal{D}(\phi) \subset T^2 \setminus (\alpha \cup \beta)$ and multiplicities $n_z(\phi), n_w(\phi)$ at the basepoints, and it contributes to $\widehat{\partial}$ only when $\mu(\phi) = 1$ and $n_z(\phi) = n_w(\phi) = 0$.



Corollary 3.1.3 (Vanishing differential). *Every Maslov index one domain covers a basepoint, so $\widehat{\partial} = 0$ and $\text{rk}(\widehat{HFK}(K)) = p$.*

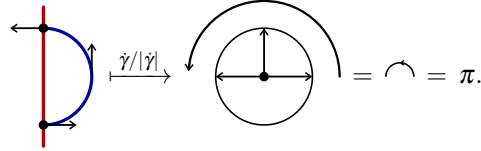
Proof.

$$\begin{aligned} \phi \in \pi_2(\mathbf{x}, \mathbf{y}), \quad \mu(\phi) = 1 &\implies \mathcal{D}(\phi) = \mathbb{D} \xrightarrow{3.1.2} n_z(\phi) + n_w(\phi) > 0 \\ &\implies \widehat{\partial} = 0 \xrightarrow{2.2.4} \text{rk}(\widehat{HFK}(K)) = \text{rk}(\widehat{CFK}(K)) = p \end{aligned}$$

□

3.2. Turning and coherence.

Definition 3.2.1 (Turning). For $\gamma \subset T^2$ oriented and immersed, $\vartheta(\gamma)$ is the total rotation of $\dot{\gamma}/|\dot{\gamma}|$ in the flat trivialisation of TT^2 ,

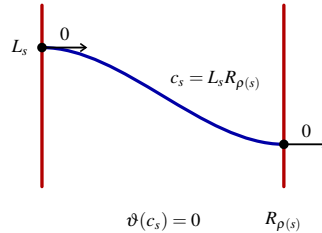


Definition 3.2.2 (Rainbow sense). Record which way each rainbow arc is traversed, up or down,

$$u_L = 1 \iff \text{blue arc up}, \quad d_L = 1 \iff \text{blue arc down}, \quad u_R = 1 \iff \text{blue arc up}, \quad d_R = 1 \iff \text{blue arc down},$$

with $d_L = 1 - u_L$ and $d_R = 1 - u_R$.

A vertical arc leaves one wall and meets the other with the same tangent direction, so it carries no turning at all.



Lemma 3.2.3 (Turning of the arcs). *Only the two rainbow arcs turn, and each by $\pm\pi$,*

$$\vartheta(c_s) = 0, \quad \vartheta(L_1 L_2) = \pi(u_L - d_L), \quad \vartheta(R_v R_{v+1}) = -\pi(u_R - d_R).$$

Proof.

$$\begin{aligned} c_s \text{ x-monotone}, \quad \dot{c}_s(0) = \dot{c}_s(1) &\implies \vartheta(c_s) = 0 \xrightarrow{9} \vartheta(\text{blue arc up}) = -\pi, \quad \vartheta(\text{blue arc down}) = +\pi \\ u \mapsto 1 - u &\implies \vartheta \mapsto -\vartheta \end{aligned}$$

□

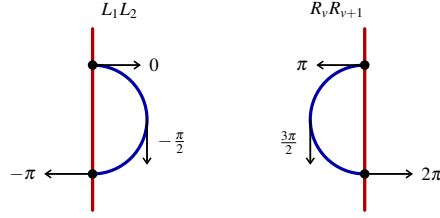


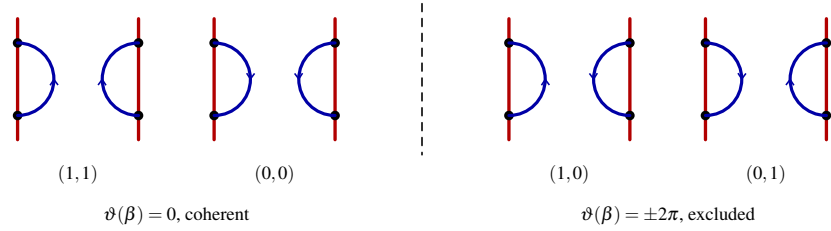
FIGURE 9. $\arg(\dot{\gamma})$ at the feet and apex of each rainbow arc, giving $\vartheta = \mp\pi$.

Proposition 3.2.4 (Turning balance). *The two rainbow arcs cancel, $u_L = u_R$ and $d_L = d_R$.*

Proof.

$$\begin{aligned} \beta \text{ embedded essential} &\implies \beta \simeq \text{geodesic} \implies \vartheta(\beta) = 0 \\ \vartheta(\beta) &= \sum_{s=3}^p \vartheta(c_s) + \vartheta(L_1L_2) + \vartheta(R_vR_{v+1}) \xrightarrow{3.2.3} \vartheta(\beta) = \pi[(u_L - d_L) - (u_R - d_R)] \\ &\implies 2u_L - 1 = 2u_R - 1 \implies u_L = u_R \end{aligned}$$

□



Corollary 3.2.5 (Shared sense). *Both rainbow arcs are traversed the same way, $\curvearrowright \curvearrowright$ and never $\curvearrowright \curvearrowleft$, so $(u_L, u_R) \in \{(0,0), (1,1)\}$.*

Proof.

$$u_L + d_L = u_R + d_R = 1 \xrightarrow{3.2.4} u_L = u_R \implies (u_L, u_R) \in \{(0,0), (1,1)\} \xLeftrightarrow{3.2.2} \curvearrowright \curvearrowright$$

□

Corollary 3.2.6 (Coherence). *$D[p, 1, b, r]$ is coherent.*

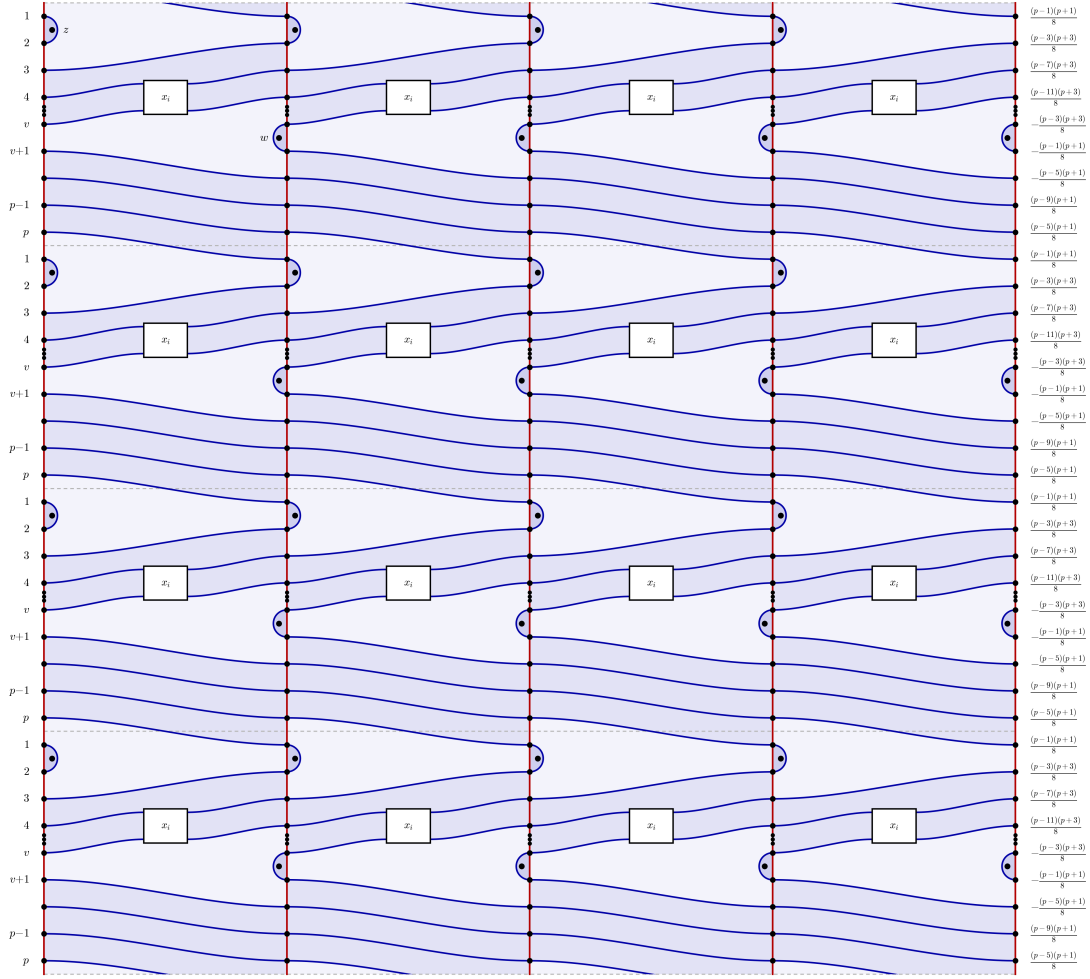
Proof.

$$\curvearrowright \curvearrowright \xrightarrow{3.2.5} \partial \curvearrowright_z, \partial \curvearrowleft_w \text{ cooriented} \xLeftrightarrow{2.3.3} D \text{ coherent}$$

□

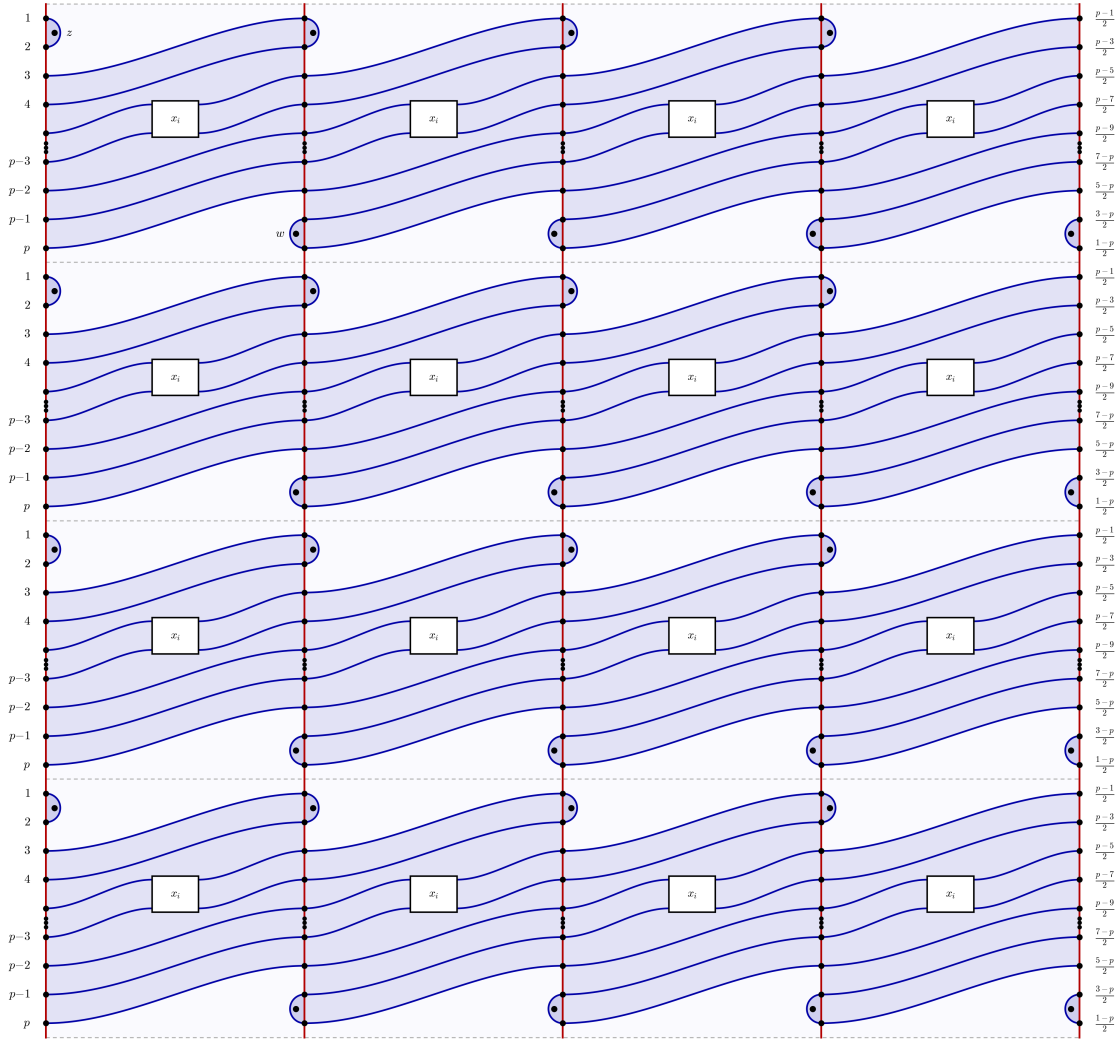
3.3. The region dichotomy.

Remark 3.3.1 (Region count, [6]). *$D[p, 1, b, r]$ has exactly two disks $\curvearrowright_z, \curvearrowleft_w$, and either one $\curvearrowright \curvearrowright$ or two $\curvearrowright \curvearrowright$, never both.*



$$A(x_i) - A(x_j) = n_z(D) - n_w(D)$$

FIGURE 10. The cover of $D[p, 1, \frac{p-3}{2}, 2]$, tinted by Definition 2.3.1, with two .



$$A(x_i) - A(x_j) = n_z(D) - n_w(D)$$

FIGURE 11. The cover of $D[p, 1, p-2, 1] = T(2, p)$, the other case of Remark 3.3.1, with one .

4. PROOF OF THEOREM A

Theorem 4.0.1 (One-rainbow coherence). *Let $[p, 1, b, r]$ be S^3 -admissible in the sense of Definition 2.2.3 and present $K \subset S^3$. Then*

- (1) $\text{rk}(\widehat{\text{HFK}}(K)) = p$, with $A_1 > \dots > A_p$ pairwise distinct.

Proof.

$$K \text{ an L-space knot} \xrightarrow{2.5.2} \text{rk}(\widehat{\text{HFK}}(K)) = 1 \text{ in each } A_j \xrightarrow{3.1.3} A_1 > \dots > A_p$$

□

- (2) $\Delta_K(t) = \sum_{j=1}^p (-1)^{j+1} t^{A_j}$, so K satisfies the Alexander criterion.

Proof.

$$A_1 > \dots > A_p \xrightarrow{2.5.2} \Delta_K(t) = \sum_{j=1}^p (-1)^{j+1} t^{A_j} \xleftrightarrow{2.5.3} \text{Alexander criterion}$$

(3) the generator of topmost Alexander grading is x_z for $e > 0$ and x_w for $e < 0$.

Proof.

$$n_z, n_w \text{ supported on } \mathbb{D}_z, \mathbb{C}_w \xrightarrow{3.1.2} A_1 \in \{A(x_z), A(x_w)\}$$

$$\text{sgn}(A(x_z) - A(x_w)) = \text{sgn}(e) \implies A_1 = A(x_z) \text{ for } e > 0, \quad A_1 = A(x_w) \text{ for } e < 0.$$

(4) K is an L-space knot.

Proof.

$$\text{admissible} \xrightarrow{3.2.6} \mathbb{D} \mathbb{C} \xrightarrow{2.3.3} D \text{ coherent} \xrightarrow{2.5.1} K \text{ an L-space knot}$$

Figure 12 is the configuration Corollary 3.2.5 excludes.

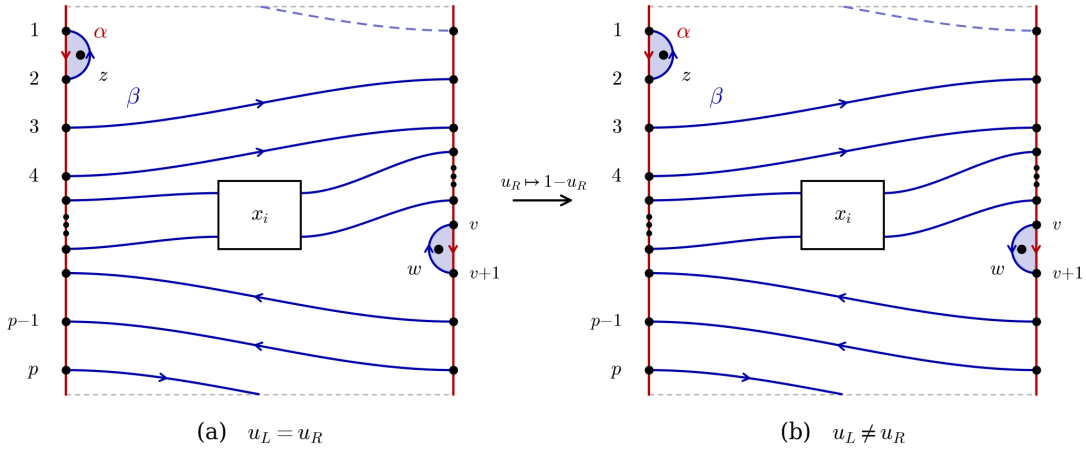
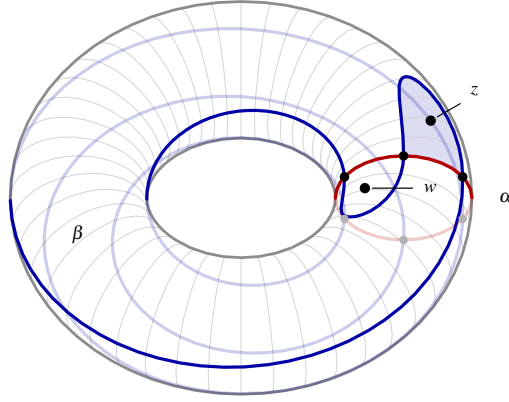


FIGURE 12. Applying $u_R \mapsto 1 - u_R$ to $\mathbb{D} \mathbb{C}$ gives $\mathbb{D} \mathbb{C}$, where $\vartheta(\beta) = \pm 2\pi$, which Corollary 3.2.6 rules out.

5. THE KERNEL

The invariant is a quotient of a kernel, $\widehat{HFK}(K) = \ker \widehat{\partial} / \text{im } \widehat{\partial}$, with $\widehat{\partial}$ the bigon count of [9, 12]. Two classical facts empty that count: on a coherent reduced (1, 1)-diagram every bigon covers a basepoint [7], and an L-space knot carries rank one in each occupied Alexander grading [10, 12].



$$n_z(\varphi) = 1$$

FIGURE 13. The fact of [7]: on a coherent reduced diagram every embedded bigon φ covers a basepoint, so the shaded rainbow bigon has $n_z(\varphi) = 1$ and contributes nothing to $\widehat{\partial}$.

Convention 5.0.1 (Matrix conventions). $\mathbb{M}_{N \times N}(\mathbb{F})$ is the algebra of $N \times N$ matrices over a field $\mathbb{F}$, with identity $\mathbf{I}_N \in \mathbb{M}_{N \times N}(\mathbb{F})$, $(\mathbf{I}_N)_{mn} = 1$ for $m = n$ and 0 otherwise. Vectors are bold columns indexed from 0, a bold letter being the column of values of its light namesake, $\mathbf{x} = (x_0, \dots, x_{N-1})^\top \in \mathbb{F}^N$, with standard basis $\mathbf{e}_n$ and $\delta_0 = \mathbf{e}_0$. For $M \in \mathbb{M}_{N \times N}(\mathbb{F})$ the row $M_{m\bullet} = (M_{m0}, \dots, M_{m,N-1})$ meets a column in a dot product,

$$(M \cdot \mathbf{x})_m = M_{m\bullet} \cdot \mathbf{x} = \sum_{n=0}^{N-1} M_{mn} \cdot x_n, \quad \ker(M) = \{\mathbf{x} \mid M \cdot \mathbf{x} = \mathbf{0}\},$$

$$\text{rk}(M) = N - \dim \ker(M), \quad \text{spec}(M) = \{\lambda \mid \det(M - \lambda \mathbf{I}_N) = 0\},$$

$$\text{GL}_N(\mathbb{F}) = \{M \in \mathbb{M}_{N \times N}(\mathbb{F}) \mid \det(M) \neq 0\};$$

M is *lower unitriangular* when $M_{nn} = 1$ and $M_{mn} = 0$ for $n > m$; these form a subgroup of $\text{GL}_N(\mathbb{F})$, with $\det(M) = 1$.

Convention 5.0.2 (The differential as a matrix). For $(T^2, \alpha, \beta, z, w)$ with $\alpha \cap \beta = \{x_1, \dots, x_p\}$,

$$\widehat{CFK}(K) = \bigoplus_{k=1}^p \mathbb{F} \cdot x_k \cong \mathbb{F}^p, \quad x_k \mapsto \mathbf{e}_{k-1}, \quad \widehat{\partial} \in \mathbb{M}_{p \times p}(\mathbb{F}).$$

As $\text{Sym}^1(T^2) = T^2$ [12, 8],

$$\mu(\varphi) = 1 \implies \mathcal{D}(\varphi) = \mathbb{D}, \quad \#(\mathcal{M}(\varphi)/\mathbb{R}) = 1,$$

$$\widehat{\partial}_{jk} = \#\{\varphi \in \pi_2(x_k, x_j) \mid \mathcal{D}(\varphi) = \mathbb{D}, n_z(\varphi) = n_w(\varphi) = 0\} \in \mathbb{F} = \mathbb{Z}/2.$$

So $\widehat{\partial}$ is the adjacency matrix of the empty-bigon relation: column k is the list of empty bigons leaving x_k ,

$$\widehat{\partial} \cdot \mathbf{e}_{k-1} = \sum_j \widehat{\partial}_{jk} \mathbf{e}_{j-1}, \quad \widehat{\partial} = \mathbf{0} \iff \text{every bigon of } T^2 \setminus (\alpha \cup \beta) \text{ carries } z \text{ or } w.$$

The bridge runs in both directions. A difference operator of calculus is a matrix through the shift $S \in \mathbb{M}_{N \times N}(\mathbb{F})$, $S_{mn} = 1$ for $n = m - 1$ and 0 otherwise, and a power of the shift is a single block,

$$\nabla_c f(u) = f(u) - f(u - c) \longleftrightarrow \mathbf{I}_N - S^c \longleftrightarrow \mathbf{I}_N - \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{I}_{N-c} & \mathbf{0} \end{bmatrix},$$

calculus on the left, $\mathbb{M}_{N \times N}(\mathbb{F})$ in the middle, 2×2 block form on the right; Proposition 6.0.1 makes the first correspondence a ring isomorphism and §6.4 computes with the second.

Example 5.0.3 (The zero differential). $[7, 1, 2, 2] \in \mathcal{C}_7$ has exactly two bigons, both occupied,

$$\{\mathcal{D}(\varphi) \mid \mu(\varphi) = 1\} = \{\mathbb{D}_z, \mathbb{D}_w\} \xrightarrow{5.0.2} \widehat{\partial}_{jk} = \#\emptyset = 0 \quad \text{for all } j, k,$$

$$\widehat{\partial} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \in \mathbb{M}_{7 \times 7}(\mathbb{F}) \xrightarrow{3.1.3} \ker(\widehat{\partial}) = \mathbb{F}^7, \quad \text{im } \widehat{\partial} = 0$$

$$\xrightarrow{5.0.1} \text{rk}(\widehat{\partial}) = 7 - 7 = 0, \quad \det(\widehat{\partial} - \lambda \mathbf{I}_7) = (-\lambda)^7, \quad \text{spec}(\widehat{\partial}) = \{0\}$$

$$\xrightarrow{[9, 7]} \widehat{HFK}(K) = \ker \widehat{\partial} / \text{im } \widehat{\partial} = \mathbb{F}^7 / 0 = \mathbb{F}^7.$$

All content therefore sits in the grading A . Its values reduce to the staircase $\mathbb{L}_{\square}$ of §6, where the matrices return as difference operators with kernel 0, the opposite spectrum (§6.4); the assignment $k \mapsto A(x_k)$ is governed by a rotation and remains open (Conjecture 6.5.1).

5.1. The census.

Definition 5.1.1 (Census).

$$\mathcal{C}_p = \{ [p, 1, b, r] \mid \text{diagram} \}, \quad \mathcal{C} = \bigcup_p \mathcal{C}_p,$$

the S^3 -admissible tuples of Definition 2.2.3.

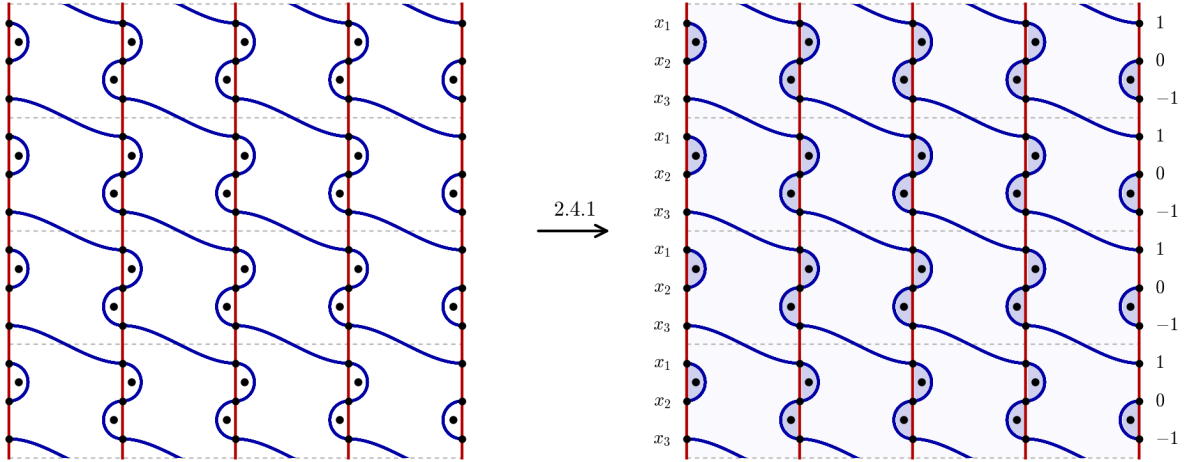
Remark 5.1.2 (Growth). $|\mathcal{C}_p| = 4, 6, 8, 8, 10, 12, 14, 16$ for $p = 3, 5, \dots, 17$, and each p contributes $T(2, p)$ and $T(\frac{p+1}{2}, \frac{p+3}{2})$, so $\mathcal{C}$ presents infinitely many knot types.

Proof.

$$\{T(2, p), T(\frac{p+1}{2}, \frac{p+3}{2})\} \xrightarrow{5.5.3, 5.5.4} \mathcal{C}_p, \quad g = \frac{p^2-1}{8} < \frac{(p+2)^2-1}{8}$$

□

The two smallest members are the diagrams of Figure 1 and Figure 2; their covers compute the gradings cell by cell.



$$(a) \quad \pi^{-1}(\alpha \cup \beta) \subset \mathbb{R}^2, \quad \pi_1(T^2) \cong \mathbb{Z}^2 = \langle T_u, T_v \rangle$$

$$(b) \quad A(x_i) - A(x_j) = n_z(D) - n_w(D)$$

FIGURE 14. The cover of $D[3, 1, 1, 1] = T(2, 3)$, given by Definition 2.3.1. Thus, $A(x_i) - A(x_j) = n_z(D) - n_w(D)$ gives $A = (1, 0, -1)$.

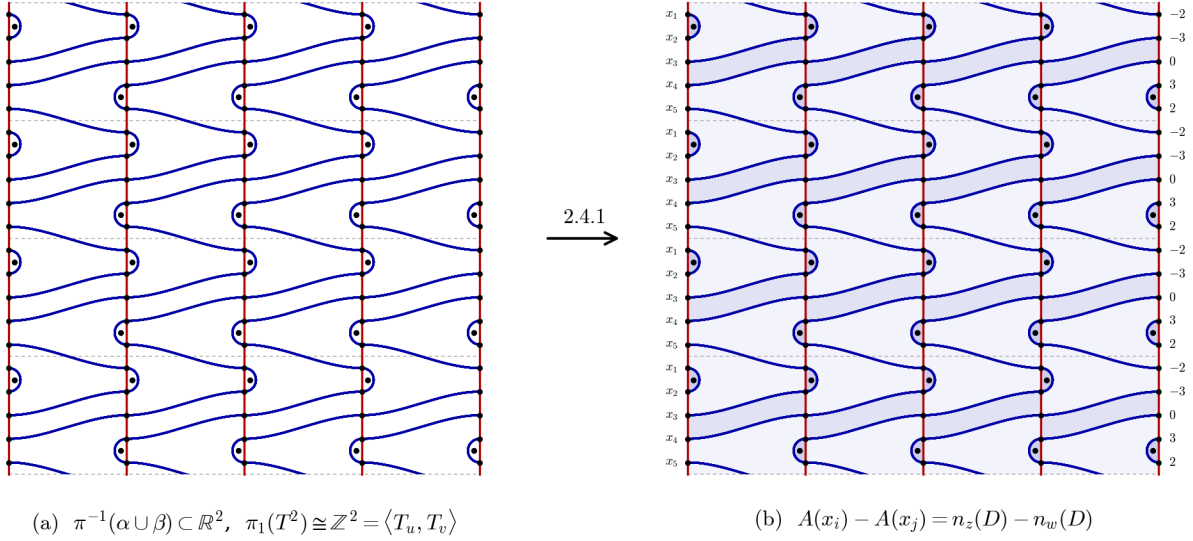


FIGURE 15. The cover of $D[5, 1, 2, 2] = T(3, 4)$, read the same way, giving $A = (3, 2, 0, -2, -3)$.

5.2. A worked tuple.

Example 5.2.1 (The worked tuple). Throughout, $D[7, 1, 2, 2]$ presents $T(4, 5)$, with $g = 6$, $S = \langle 4, 5 \rangle$ and

$$(A(x_1), \dots, A(x_7)) = (6, 5, 0, -5, -6, -2, 2),$$

read cell by cell as in Figure 15 and placed by (A, M) in Figure 16.

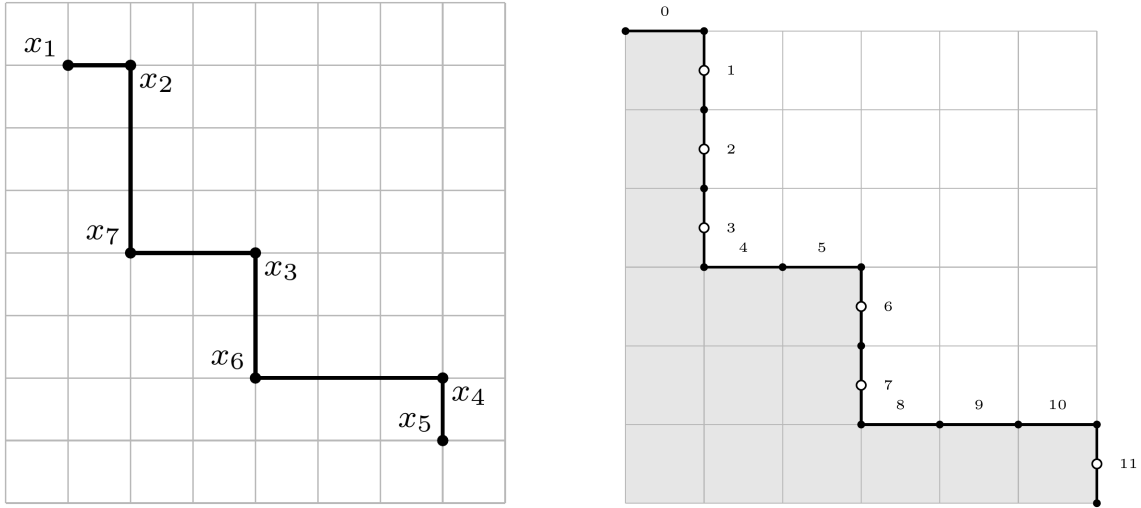


FIGURE 16. The complex of $[7, 1, 2, 2]$ placed by (A, M) , arrowless by Corollary 3.1.3, and the same path read as the staircase $\sqcup_{\sqcup}$ of $S = \langle 4, 5 \rangle$, gaps $\circ$ ringed.

5.3. Perfect diagrams.

Definition 5.3.1 (Perfect diagram). A reduced $(1, 1)$ -diagram is *perfect* when $\widehat{\partial} = 0$ and $A(x_i) \neq A(x_j)$ for $i \neq j$.

Proposition 5.3.2 (Support). *Every member of $\mathcal{C}$ is perfect, with*

$$\{A(x_1), \dots, A(x_p)\} = \text{Supp}(\Delta_K)$$

and every coefficient of Δ_K equal to ± 1 .

Proof.

$$\widehat{\partial} = 0 \xrightarrow{3.1.3} \widehat{HFK}(K) \cong \bigoplus_k \mathbb{F}_{A(x_k)} \xrightarrow{4.0.1} A(x_i) \neq A(x_j) \ (i \neq j) \xrightarrow{2.5.2} \text{Supp}(\Delta_K) = \{A(x_k)\}$$

□

5.4. The difference word.

Definition 5.4.1 (Difference word). Reading indices in $\mathbb{Z}/p$,

$$\mathbf{d} = (d_1, \dots, d_p), \quad d_k = A(x_k) - A(x_{k+1}), \quad \sum_k d_k = 0,$$

and $\mathbf{d}$ determines A up to the constant fixed by Lemma 2.4.2.

Lemma 5.4.2 (Rainbow steps). $d_1 = d_v = 1$ for every tuple in $\mathcal{C}$.

Proof.

$$\begin{aligned} x_1, x_2 \in \partial \mathbb{D}_z &\xrightarrow{2.1.6} (n_z, n_w)(\mathbb{D}_z) = (1, 0) \xrightarrow{[9]} d_1 = n_z - n_w = 1 \\ x_v, x_{v+1} \in \partial \mathbb{C}_w &\xrightarrow{2.4.2} (n_z, n_w)(\mathbb{C}_w) = (0, 1) \xrightarrow{2.4.2} d_v = 1 \end{aligned}$$

□

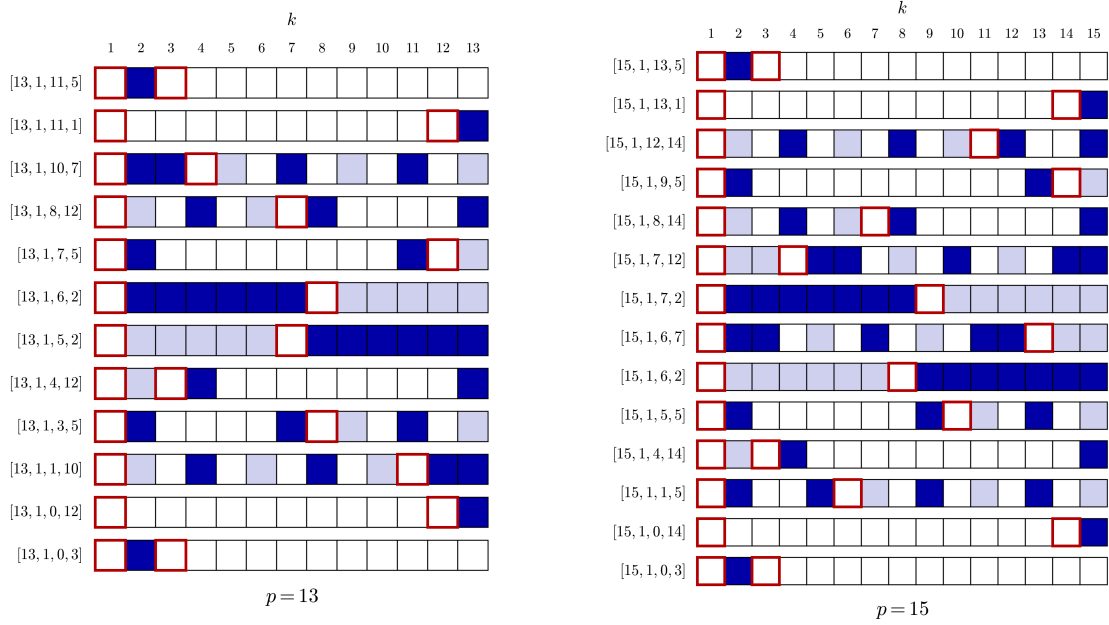


FIGURE 17. $\mathbf{d}$ for every tuple of $\mathcal{C}_{13}$ and $\mathcal{C}_{15}$, one strip each, k across; two-run rows read as blocks, the rest interleave.

Conjecture 5.4.3 (Three letters). *For every tuple in $\mathcal{C}$ there is a single $c \geq 2$ with*

$$\{d_1, \dots, d_p\} \subseteq \{1, -c, c+1\}.$$

Conjecture 5.4.4 (Three gaps). *For each letter u with $P_u = \{k \in \mathbb{Z}/p \mid d_k = u\}$, the cyclic differences of consecutive elements of P_u take values*

$$\{g_1\}, \quad \{g_1, g_2\} \quad \text{or} \quad \{g_1, g_2, g_1 + g_2\}.$$

Both hold for the 62 tuples with $p \leq 15$; the single departure from the letter set is $\{1, -4, 4\}$, at $[7, 1, 1, 5]$ and $[7, 1, 4, 6]$.

5.5. Two closed forms.

Definition 5.5.1 (Two-run word). $\mathbf{d}$ is *two-run* when $d_1 = d_v = 1$, $d_k = u$ for $2 \leq k \leq v-1$ and $d_k = D$ for $v+1 \leq k \leq p$.

Proposition 5.5.2 (Determination). *A two-run word is determined by p , v and g ,*

$$(v-2)u = 2g-2, \quad (p-v)D = -2g.$$

Proof.

$$\begin{aligned} \sum_k d_k &= 0 \xrightarrow{5.5.1} 2 + (v-2)u + (p-v)D = 0 \\ A(x_1) &= g, \quad A(x_{v+1}) = g-2 - (v-2)u \xrightarrow{5.3.2} A(x_{v+1}) = -g \implies (v-2)u = 2g-2 \end{aligned}$$

□

Corollary 5.5.3 (Consecutive torus knots). *For $[p, 1, \frac{p-3}{2}, 2]$, $v = \frac{p+1}{2}$, $K = T(v, v+1)$, $g = \frac{p^2-1}{8}$, and*

$$\begin{aligned} A(x_1) &= \frac{(p-1)(p+1)}{8}, \quad A(x_k) = \frac{(p-4k+5)(p+3)}{8} \quad (2 \leq k \leq v), \\ A(x_k) &= \frac{(4k-3p-5)(p+1)}{8} \quad (v < k \leq p). \end{aligned}$$

This is Figure 10.

Proof.

$$g = \frac{p^2-1}{8}, \quad v = \frac{p+1}{2} \xrightarrow{5.5.2} u = \frac{p+3}{2}, \quad D = -\frac{p+1}{2} \xrightarrow{5.4.1} A(x_k) = g - \sum_{i < k} d_i$$

□

Corollary 5.5.4 (Two-strand torus knots). *For $[p, 1, p-2, 1]$, $v = p-1$, $K = T(2, p)$, $S = \langle 2, p \rangle$, $g = \frac{p-1}{2}$, and*

$$A(x_k) = \frac{p+1-2k}{2}, \quad 1 \leq k \leq p.$$

This is Figure 11.

Proof.

$$g = \frac{p-1}{2}, \quad v = p-1 \xrightarrow{5.5.2} u = 1, \quad D = -(p-1) \implies A(x_k) = \frac{p+1-2k}{2}$$

□

6. THE BOX SPLINE

By Proposition 5.3.2 the exponents of Δ_K are cut out by $S_K \subset \mathbb{Z}_{\geq 0}$. Four objects carry the same data, two discrete and two continuous, by Definition 2.6.2 and Definition 2.6.1,

$$d(n; a, b) = |\{(i, j) \in \mathbb{Z}_{\geq 0}^2 \mid ia + jb = n\}|, \quad n \in S \iff d(n; a, b) \geq 1,$$

$$T(u) = \frac{u_+}{ab}, \quad B_{(a,b)} = \nabla_a \nabla_b T, \quad \nabla_c f(u) = f(u) - f(u-c),$$

one operator $\nabla_a \nabla_b$ acting on both rows.

$$\begin{array}{ccc}
T(u) = u_+ / ab & \xrightarrow[\text{[3]}]{\nabla_a \nabla_b} & B_{(2,3)} \\
\downarrow \text{[11]} & & \downarrow \text{[15]} \\
d(n; 2, 3) & \xrightarrow[\text{[4]}]{\nabla_a \nabla_b} & \delta_0 \\
(1, 0, 1, 1, 1, 1, 2, 1, \dots) & & (1, 0, 0, 0, \dots)
\end{array}$$

FIGURE 18. The square for $(a, b) = (2, 3)$. Across, the second difference $\nabla_a \nabla_b$ carries the truncated power to the box spline [3] and the denumerant to δ_0 [4]. Down, the continuous objects pass to their lattice counts [15, 11]; the gap $1 \notin S = \langle 2, 3 \rangle$ is the one zero.

The sample window is that of Example 5.2.1: $[7, 1, 2, 2]$, $K = T(4, 5)$, $g = 6$, $ab = 20$. Every operator below is a polynomial in the shift, and matrix computation is polynomial multiplication through one dictionary, under Convention 5.0.1.

Proposition 6.0.1 (The symbol dictionary). *On $\mathbb{F}^N$ with shift S , $S_{mn} = 1$ for $n = m - 1$, the map $f \mapsto f(S) = \sum_{n < N} f_n \cdot S^n$ is a ring isomorphism from $\mathbb{F}[t]/(t^N)$ onto the lower-triangular Toeplitz matrices in $\mathbb{M}_{N \times N}(\mathbb{F})$ [13, 3],*

$$f(S) \cdot g(S) = (fg)(S), \quad (f(S))_{mn} = f_{m-n}, \quad f(S) = \begin{bmatrix} f(S_h) & \mathbf{0} \\ H_f & f(S_h) \end{bmatrix} \quad (N = 2h),$$

with $(H_f)_{ij} = f_{h+i-j}$ and equal diagonal blocks.

Proof.

$$(f(S) \cdot g(S))_{mn} = \sum_k (f(S))_{mk} \cdot (g(S))_{kn} = \sum_k f_{m-k} \cdot g_{k-n} = (fg)_{m-n},$$

and each block of $(f(S))_{mn} = f_{m-n}$ depends only on $m - n$, the two diagonal blocks equally. $\square$

6.1. The associated set.

Definition 6.1.1 (Associated set, [10, 1]). For an L-space knot K ,

$$t^g \Delta_K(t) = (1 - t) \sum_{s \in S_K, s < 2g} t^s + t^{2g}, \quad S_K \supset \mathbb{Z}_{\geq 2g}, \quad \mathcal{G}_K = \mathbb{Z}_{\geq 0} \setminus S_K,$$

so the partial sums of the coefficients of $t^g \Delta_K$ are the indicator of S_K . For $K = T(a, b)$ this is Definition 2.6.1 by Lemma 2.6.3.

Example 6.1.2 (Partial summation as a symbol). By Proposition 6.0.1, partial summation on $n = 0, \dots, 12$ is the geometric symbol $P = (1 + t + \dots + t^{12})(S)$, inverse to $1 - t$, applied to the coefficient polynomial $c(t) = t^6 \Delta_{T(4,5)} = 1 - t + t^4 - t^6 + t^8 - t^{11} + t^{12}$ of Definition 6.1.1, every cross term expanded. Explicitly,

$$\begin{aligned}
(1 - t) \cdot (1 + t^4 + t^5 + t^8 + t^9 + t^{10}) &= 1 + t^4 + t^5 + t^8 + t^9 + t^{10} - t - t^5 - t^6 - t^9 - t^{10} - t^{11} \\
&= 1 - t + t^4 - t^6 + t^8 - t^{11} = c(t) - t^{12} \\
\implies P \cdot \mathbf{c} &\longleftrightarrow \frac{c(t)}{1 - t} \equiv 1 + t^4 + t^5 + t^8 + t^9 + t^{10} + t^{12} \longleftrightarrow (1, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1)^\top = \mathbf{1}_S \\
&\xrightarrow{2.6.3} \mathcal{G} = \{1, 2, 3, 6, 7, 11\}, \quad |\mathcal{G}| = 6 = g.
\end{aligned}$$

Proposition 6.1.3 (Census semigroups). *For every tuple in $\mathcal{C}$ with $p \leq 13$, $|\mathcal{G}_K| = g$. For 40 of the 48 tuples $S_K + S_K \subset S_K$, and then $S_K = \langle a, b \rangle$ with $g = \frac{(a-1)(b-1)}{2}$ and $K = T(a, b)$. The remaining eight,*

$$[9, 1, 3, 5], [9, 1, 4, 8], [11, 1, 4, 10], [11, 1, 5, 5], [13, 1, 3, 5], [13, 1, 4, 12], [13, 1, 7, 5], [13, 1, 8, 12],$$

are L-space knots that are not algebraic [1].

Proof.

$$\Delta_K \text{ alternating} \xrightarrow{2.5.2} P \cdot \mathbf{c} \in \{0, 1\}^{2g+1} \implies |\mathcal{G}_K| = \deg(t^g \Delta_K) - g = g,$$

and $S_K + S_K \subset S_K$ is a finite check over the 48 tuples of Definition 5.1.1 (Remark 5.1.2). $\square$

Lemma 6.1.4 (Gap symmetry, [15]). *For $S = \langle a, b \rangle$,*

$$n \in \mathcal{G} \iff 2g - 1 - n \in S, \quad |\mathcal{G}| = g,$$

equivalently $\Delta_K(t^{-1}) = \Delta_K(t)$ for $K = T(a, b)$.

Proof.

$$\mathcal{G} = \{1, 2, 3, 6, 7, 11\} \implies 2g - 1 - \mathcal{G} = \{10, 9, 8, 5, 4, 0\} \subset S \xrightarrow{2.6.3} \Delta_K(t^{-1}) = \Delta_K(t),$$

and in general [15]. □

Example 6.1.5 (Gap symmetry as reversal). The exchange matrix $J \in \mathbb{M}_{12 \times 12}(\mathbb{F})$, $J_{mn} = 1$ for $m + n = 2g - 1 = 11$ and $J_{mn} = 0$ otherwise, so $(J \cdot \mathbf{x})_m = x_{11-m}$, is the elliptic involution $\iota(u) = -u$ of the Heegaard torus in coordinates (Lemma 2.4.2, [9, 12]),

$$J = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad J^2 = \mathbf{I}_{12},$$

$$J \cdot S \cdot J = S^\top \xrightarrow{6.0.1} J \cdot f(S) \cdot J = f(S)^\top,$$

$$J \cdot \mathbf{1}_S = J \cdot (1, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0)^\top = (0, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0, 1)^\top = \mathbf{1} - \mathbf{1}_S \xrightarrow{6.1.4} \Delta_K(t^{-1}) = \Delta_K(t).$$

6.2. The partition function.

Proposition 6.2.1 (Generating identities, [13, 4]). *For $\gcd(\{a, b\}) = 1$,*

$$\sum_{n \geq 0} d(n; a, b) t^n = \frac{1}{(1 - t^a)(1 - t^b)}, \quad (1 - t^{ab}) \sum_{n \geq 0} d(n; a, b) t^n = \sum_{s \in S} t^s,$$

$$\Delta_{T(a, b)}(t) = t^{-g}(1 - t)(1 - t^{ab}) \sum_{n \geq 0} d(n; a, b) t^n.$$

Proof.

$$\sum_{n \geq 0} d(n; a, b) t^n = \sum_{i, j \geq 0} t^{ia + jb} = \frac{1}{(1 - t^a)(1 - t^b)} \xrightarrow{2.6.1} (1 - t^{ab}) \sum_n d t^n = \sum_{s \in S} t^s \xrightarrow{2.6.3} \Delta_{T(a, b)}(t) = t^{-g}(1 - t)(1 - t^{ab}) \sum_n d t^n$$

□

Proposition 6.2.2 (Closed form, [11]). *With $a^{-1}a \equiv 1 \pmod{b}$ and $b^{-1}b \equiv 1 \pmod{a}$,*

$$d(n; a, b) = \frac{n}{ab} - \left\{ \frac{b^{-1}n}{a} \right\} - \left\{ \frac{a^{-1}n}{b} \right\} + 1, \quad \mathcal{G} = \{n \mid d(n; a, b) = 0\}.$$

Proof. [11]; $d(n) = 0 \iff n \notin S$ is Definition 2.6.1. □

Example 6.2.3 (The count at $n = 4, 11$). $(a, b) = (4, 5)$, $b^{-1} = 1$, $a^{-1} = 4$, over the denominator $ab = 20$,

$$d(4) = \frac{4}{20} - \left\{ \frac{4}{4} \right\} - \left\{ \frac{16}{5} \right\} + 1 = \frac{4}{20} - \frac{0}{20} - \frac{4}{20} + \frac{20}{20} = 1, \quad d(11) = \frac{11}{20} - \frac{15}{20} - \frac{16}{20} + \frac{20}{20} = 0$$

$$\xrightarrow{6.1.4} 11 = ab - a - b = \max \mathcal{G}.$$

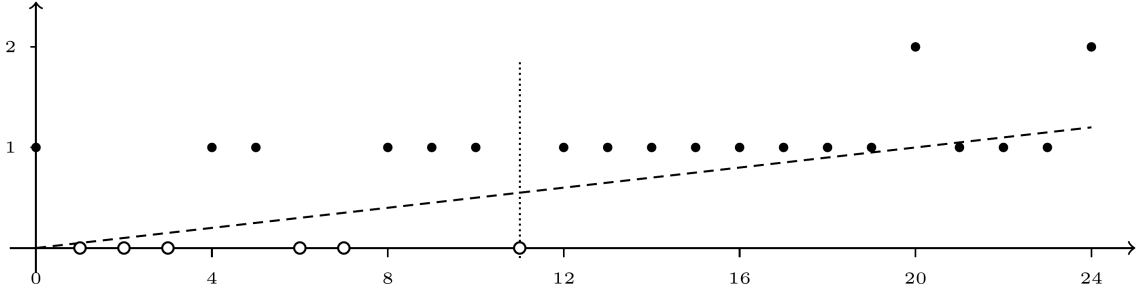


FIGURE 19. $d(n; 4, 5)$ against the line n/ab it samples; the zeros $\circ$ are exactly $\mathcal{G}$, the last at $ab - a - b = 11$.

6.3. The tent.

Lemma 6.3.1 (Tent form, [3]). *For $a < b$, $B_{(a,b)} = \triangleleft$ is piecewise linear with knots $0, a, b, a + b$,*

$$abB_{(a,b)}(u) = \begin{cases} u, & 0 \leq u \leq a, \\ a, & a \leq u \leq b, \\ a + b - u, & b \leq u \leq a + b, \\ 0, & \text{otherwise.} \end{cases}$$

Proof.

$$B_{(a,b)} \xrightarrow{2.6.2} \frac{1}{a}\chi_{[0,a]} * \frac{1}{b}\chi_{[0,b]} \implies abB_{(a,b)}(u) = \max\{0, \min(u, a, a + b - u)\}$$

□

Example 6.3.2 (The convolution). The convolution of Definition 2.6.2 is one integral [3],

$$20B_{(4,5)}(u) = \int_{\mathbb{R}} \chi_{[0,4]}(x) \chi_{[u-5,u]}(x) dx = |[0,4] \cap [u-5,u]|, \quad 20B_{(4,5)}(7) = \int_2^4 dx = 2.$$

Theorem 6.3.3 (Spline form of the Alexander polynomial). *Let $K = T(a, b)$ and $T(u) = u_+/ab$. Then*

$$\nabla_a \nabla_b T = B_{(a,b)}, \quad \nabla_a \nabla_b d = \delta_0, \quad d - T|_{\mathbb{Z}} \text{ ab-periodic}, \quad |d - T| \leq 1,$$

and

$$t^S \Delta_K(t) = (1-t)(1-t^{ab}) \sum_{n \geq 0} d(n; a, b) t^n, \quad \mathcal{G} = \{n \mid d(n; a, b) = 0\},$$

so Δ_K , S , $d(\cdot; a, b)$ and $B_{(a,b)}$ determine one another.

Proof.

$$\begin{aligned} \nabla_a \nabla_b T &\xrightarrow{6.3.1} B_{(a,b)} \\ (1-t^a)(1-t^b) \sum_n d t^n &= 1 \xLeftrightarrow{6.2.1} \nabla_a \nabla_b d = \delta_0 \iff L \cdot \mathbf{d} = \delta_0 \\ (d-T)(n) &= 1 - \left\{ \frac{b^{-1}n}{a} \right\} - \left\{ \frac{a^{-1}n}{b} \right\} \xrightarrow{6.2.2} (d-T)(n+ab) = (d-T)(n), \quad |d-T| \leq 1 \\ &\xrightarrow{6.2.1} t^S \Delta_K(t) = (1-t)(1-t^{ab}) \sum_{n \geq 0} d(n; a, b) t^n, \quad \mathcal{G} = d^{-1}(0) \end{aligned}$$

□

6.4. The difference matrix. $\deg((1-t^4)(1-t^5)) = 9$, so $N = 10$ is the shortest window on which $\nabla_4 \nabla_5$ of Theorem 6.3.3 is a full matrix $L = \ell(S)$, symbol

$$\ell = (1-t^4) \cdot (1-t^5) = 1 - t^4 - t^5 + t^9,$$

shown once in full and once in the blocks of Proposition 6.0.1,

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} A & \mathbf{0} \\ B & A \end{bmatrix},$$

$$A = \ell(S_5) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = H_\ell = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 \\ 1 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

Example 6.4.1 (The two transports). Blockwise on $\mathbf{T} = 20T|_{\mathbb{Z}} = (0, \dots, 9)^\top$ and on the partition function $\mathbf{d}$ of Proposition 6.2.1 (not the word of Definition 5.4.1),

$$L \cdot \mathbf{T} = \begin{bmatrix} A \cdot \mathbf{T}^0 \\ B \cdot \mathbf{T}^0 + A \cdot \mathbf{T}^1 \end{bmatrix}, \quad A \cdot \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ -3 \\ -5 \\ -7 \\ -4 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\xrightarrow{6.3.1} L \cdot \mathbf{T} = \mathbf{B} = 20B_{(4,5)}|_{\mathbb{Z}};$$

$$A \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow{6.3.3} L \cdot \mathbf{d} = \delta_0, \quad \nabla_4 \nabla_5 d = \delta_0.$$

Proposition 6.4.2 (Unimodularity and spectrum). L is lower unitriangular, so

$$\det(L - \lambda \mathbf{I}) = (1 - \lambda)^{10}, \quad \det(L) = 1, \quad \dim \ker(L) = 0, \quad \text{rk}(L) = 10,$$

the one eigenvalue of L is 1 with multiplicity 10, and $\mathbf{d}$ is the unique solution of $L \cdot \mathbf{d} = \delta_0$.

Proof.

$$\det(L - \lambda \mathbf{I}) = \det(A - \lambda \mathbf{I}_5)^2 = ((\ell_0 - \lambda)^5)^2 = (1 - \lambda)^{10} \xrightarrow{\lambda=0} \det(L) = 1 \implies \ker(L) = 0.$$

□

Proposition 6.4.3 (Unipotent inverse). Let $N = S^4 + S^5 - S^9$, so $L = \mathbf{I} - N$. Then

$$N^3 = 0, \quad L^{-1} = \mathbf{I} + N + N^2 = \sum_{s \in S, s < 10} S^s, \quad (L^{-1})_{mn} = d(m-n), \quad L^{-1} \cdot \delta_0 = \mathbf{d}.$$

Proof. With $n = t^4 + t^5 - t^9$, multiplying symbols modulo t^{10} by Proposition 6.0.1,

$$\begin{aligned} n^2 &= t^8 + t^9 + t^9 = t^8 + 2t^9, & n^3 &\equiv 0 \quad (\text{each factor raises the valuation by } \geq 4, 3 \cdot 4 > 9), \\ \implies \ell \cdot (1 + n + n^2) &= 1 - n^3 \equiv 1 \xrightarrow{6.2.1} (L^{-1})_{mn} = d(m - n). \end{aligned}$$

□

Example 6.4.4 (The inverse). By Proposition 6.4.3 and Proposition 6.0.1, L^{-1} is the symbol of the semigroup, $(L^{-1})_{mn} = d(m - n)$ is Toeplitz, so L^{-1} is a convolution, i.e. a spline [3, 4], and its column 0 is $\mathbf{d}$, the staircase of Figure 16 read off the inverse. Explicitly,

$$\begin{aligned} 1 + n + n^2 &= 1 + (t^4 + t^5 - t^9) + (t^8 + 2t^9) = 1 + t^4 + t^5 + t^8 + t^9 - t^4 - t^8 - t^9 - t^5 - t^9 + t^9 = \sum_{s \in S, s < 10} t^s \\ \implies \ell \cdot (1 + t^4 + t^5 + t^8 + t^9) &= 1 + t^4 + t^5 + t^8 + t^9 - t^4 - t^8 - t^9 - t^5 - t^9 + t^9 = 1 \pmod{t^{10}}; \\ \ell &\equiv 1 - t^4 \pmod{t^5}, \quad (S_5^4)^2 = \mathbf{0} \implies A^{-1} = (1 + t^4)(S_5) = \mathbf{I}_5 + S_5^4 \\ L^{-1} &= \begin{bmatrix} A & \mathbf{0} \\ B & A \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & \mathbf{0} \\ -A^{-1} \cdot B \cdot A^{-1} & A^{-1} \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}, \\ B \cdot A^{-1} &= \begin{bmatrix} -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 \\ 1 & 0 & 0 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ -1 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}, \\ -A^{-1} \cdot (B \cdot A^{-1}) &= - \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ -1 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix} \\ \implies L \cdot L^{-1} &= \begin{bmatrix} A \cdot A^{-1} & \mathbf{0} \\ B \cdot A^{-1} - A \cdot A^{-1} \cdot B \cdot A^{-1} & A \cdot A^{-1} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_5 & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_5 \end{bmatrix} = \mathbf{I}_{10} \\ \implies L^{-1} &= (1 + t^4 + t^5 + t^8 + t^9)(S) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \\ \implies L^{-1} \cdot \delta_0 &= \mathbf{d}. \end{aligned}$$

Against $\widehat{\partial} = 0_{7 \times 7}$ of Example 5.0.3 the operator $\nabla_4 \nabla_5$ is the opposite,

$$\begin{aligned} \dim \ker(\widehat{\partial}) &= p = 7, & \text{rk}(\widehat{\partial}) &= 0, & \text{spec}(\widehat{\partial}) &= \{0\}, \\ \dim \ker(L) &= 0, & \text{rk}(L) &= 10, & \text{spec}(L) &= \{1\}; \end{aligned}$$

one nilpotent as a differential [9], one unipotent as an operator [3],

$$\widehat{\partial}^2 = 0, \quad (L - \mathbf{I})^3 = 0.$$

The first stores $\widehat{HFK}(K)$ in its kernel, the second determines Δ_K by inversion through Proposition 6.4.3.

Remark 6.4.5 (The chain). Figure 16 and Figure 19 carry $[7, 1, 2, 2]$ from complex to box spline, the staircase of $\widehat{CFK}(K)$ and the staircase of the monomial ideal of $k[t^4, t^5]$ being the same path $[10, 1, 4]$.

Proof. Proposition 5.3.2, Proposition 6.2.1, Theorem 6.3.3. $\square$

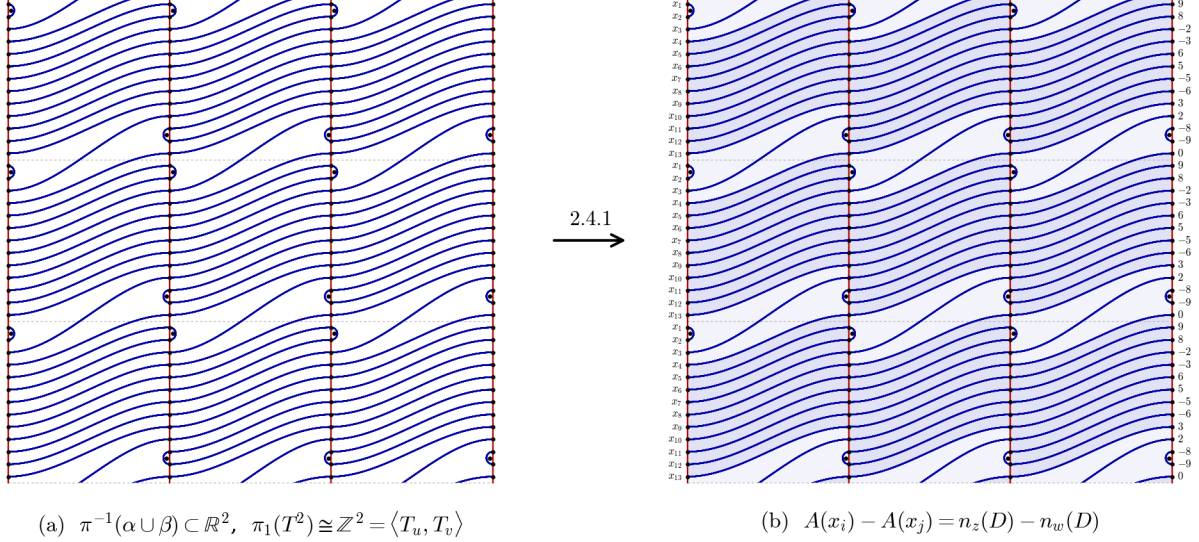


FIGURE 20. The computed cover of $[13, 1, 1, 10] = T(3, 10)$, a tuple off both branches, tinted by Definition 2.3.1; generators down the left margin, gradings down the right.

6.5. The conjecture.

Conjecture 6.5.1 (Rotation data). *For every $[p, 1, b, r] \in \mathcal{C}$ there are $c, q, s \in \mathbb{Z}$ with*

$$A(x_k) = \lambda k - c \left\lfloor \frac{kq + s}{p} \right\rfloor + \varepsilon_k, \quad \text{Supp}(\varepsilon) \subset \{1, v\}.$$

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APPENDIX A. FOUNDATIONAL TOOLS

A.1. Linear algebra over $\mathbb{F}$. Under Convention 5.0.1 a matrix is the map it induces, and its columns are where the basis goes,

$$M \cdot \mathbf{e}_n = (M_{0n}, \dots, M_{N-1,n})^\top, \quad \ker(M) = \{\mathbf{x} \mid M \cdot \mathbf{x} = \mathbf{0}\}, \quad \text{rk}(M) + \dim \ker(M) = N.$$

Example A.1.1 (The shift on $\mathbb{F}^3$).

$$S = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad S \cdot \mathbf{e}_0 = \mathbf{e}_1, \quad S \cdot \mathbf{e}_1 = \mathbf{e}_2, \quad S \cdot \mathbf{e}_2 = \mathbf{0},$$

$$S^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad S^3 = \mathbf{0}, \quad \text{rk}(S) = 2, \quad \dim \ker(S) = 1.$$

The basis slides one step and the last vector is dropped, so the kernels grow by one dimension at each power,

$$\{\mathbf{0}\} \subset \ker S = \langle \mathbf{e}_2 \rangle \subset \ker S^2 = \langle \mathbf{e}_1, \mathbf{e}_2 \rangle \subset \ker S^3 = \mathbb{F}^3,$$

a line inside a plane inside the space, and $\text{im } S = \langle \mathbf{e}_1, \mathbf{e}_2 \rangle$, $\text{spec}(S) = \{0\}$.

Example A.1.2 (Difference and summation). $L = \mathbf{I}_3 - S$ is ∇_1 , and its inverse is partial summation,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \quad L^{-1} = \mathbf{I}_3 + S + S^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad L \cdot L^{-1} = \mathbf{I}_3,$$

$$(L \cdot \mathbf{x})_m = x_m - x_{m-1}, \quad (L^{-1} \cdot \mathbf{x})_m = \sum_{n \leq m} x_n, \quad \det(L - \lambda \mathbf{I}_3) = (1 - \lambda)^3.$$

Example A.1.3 (Three null spaces). The three regimes of the paper, all on $\mathbb{F}^3$,

$$\widehat{\partial} = \mathbf{0} \implies \ker = \mathbb{F}^3 \text{ (the space),} \quad \text{rk} = 0, \quad \text{spec} = \{0\},$$

$$S \implies \ker = \langle \mathbf{e}_2 \rangle \text{ (a line),} \quad \text{rk} = 2, \quad \text{spec} = \{0\},$$

$$L = \mathbf{I}_3 - S \implies \ker = \{\mathbf{0}\} \text{ (the origin),} \quad \text{rk} = 3, \quad \text{spec} = \{1\}.$$

Example 5.0.3 and §6.4 are the two ends of that list: the differential whose null space is everything, and the difference operator whose null space is a point.

Proposition A.1.4 (Unitriangular). *Let $M \in \mathbb{M}_{N \times N}(\mathbb{F})$ be lower unitriangular in the sense of Convention 5.0.1, and $N_M = \mathbf{I}_N - M$. Then*

$$\det(M - \lambda \mathbf{I}_N) = (1 - \lambda)^N, \quad \det(M) = 1, \quad \text{spec}(M) = \{1\}, \quad (N_M)^N = \mathbf{0}.$$

Proof.

$$\begin{aligned} M_{mn} &= 0 \ (n > m), \quad M_{nn} = 1 \implies M - \lambda \mathbf{I}_N \text{ is lower triangular with diagonal } 1 - \lambda \\ &\implies \det(M - \lambda \mathbf{I}_N) = (1 - \lambda)^N \xrightarrow{\lambda=0} \det(M) = 1 \implies M \in \text{GL}_N(\mathbb{F}) \\ (N_M)_{mn} &= 0 \ (n \geq m) \implies ((N_M)^k)_{mn} = 0 \ (m - n < k) \implies (N_M)^N = \mathbf{0}. \end{aligned}$$

□

Proposition A.1.5 (Neumann inverse). *If $N^{m+1} = \mathbf{0}$ then $\mathbf{I} - N$ is invertible with*

$$(\mathbf{I} - N)^{-1} = \sum_{k=0}^m N^k.$$

Proof.

$$(\mathbf{I} - N) \sum_{k=0}^m N^k = \sum_{k=0}^m N^k - \sum_{k=1}^{m+1} N^k = \mathbf{I} - N^{m+1} = \mathbf{I},$$

and the same computation on the other side. □

Example A.1.6 (Reversing the basis). The exchange matrix reverses the order of the basis, and conjugation by it turns the shift into its transpose,

$$J = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad J \cdot \mathbf{e}_n = \mathbf{e}_{2-n}, \quad J^2 = \mathbf{I}_3, \quad J \cdot S \cdot J = S^\top,$$

$$\xrightarrow{6.0.1} J \cdot f(S) \cdot J = f(S)^\top \quad \text{for every symbol } f,$$

which is Example 6.1.5 at $N = 12$.

Example A.1.7 (Changing basis on $H_1(T^2)$). The development of Figure 1 rewrites the square's coordinates in the meridian–longitude basis,

$$H = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \quad H \cdot \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u \\ v - u \end{pmatrix}, \quad H^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \det H = 1.$$

So $H \in \text{SL}_2(\mathbb{Z})$ is an automorphism of $H_1(T^2) = \mathbb{Z}^2$; it preserves gcd, hence primitivity, and

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix} \mapsto \begin{pmatrix} 2 \\ -3 \end{pmatrix}, \quad \gcd(\{2, 3\}) = 1 \implies K = T(2, 3).$$

A.2. Numerical semigroups.

Lemma A.2.1 (Canonical form). *For $\gcd(\{a, b\}) = 1$ and $n \in \mathbb{Z}$ there are unique $i \in \{0, \dots, b-1\}$ and $j \in \mathbb{Z}$ with $n = ia + jb$, and*

$$n \in S = \langle a, b \rangle \iff j \geq 0.$$

Proof.

$$\begin{aligned} \gcd(\{a, b\}) = 1 &\implies a \in (\mathbb{Z}/b)^\times \implies \exists! i \in \{0, \dots, b-1\}, ia \equiv n \pmod{b}, \quad j = \frac{n-ia}{b} \in \mathbb{Z} \\ j \geq 0 &\implies n = ia + jb \in S \\ n = i'a + j'b, i', j' \geq 0 &\implies i' \equiv i \pmod{b} \implies i' = i + kb, k \geq 0 \\ \implies j' = \frac{n-i'a}{b} = j - ka &\implies j = j' + ka \geq 0. \end{aligned}$$

□

Theorem A.2.2 (Sylvester, [15]). *For $\gcd(\{a, b\}) = 1$ and $F = ab - a - b$, exactly one of $n \in S$, $F - n \in S$ holds for every $n \in \mathbb{Z}$; hence*

$$\mathcal{G} \subset \{0, \dots, F\}, \quad \max \mathcal{G} = F = 2g - 1, \quad |\mathcal{G}| = \frac{(a-1)(b-1)}{2} = g.$$

Proof.

$$\begin{aligned} n = ia + jb, \quad 0 \leq i \leq b-1 &\xrightarrow{A.2.1} F - n = (b-1-i)a + (-1-j)b, \quad 0 \leq b-1-i \leq b-1 \\ &\xrightarrow{A.2.1} F - n \in S \iff -1-j \geq 0 \iff j \leq -1 \iff n \notin S \\ 0 = 0 \cdot a + 0 \cdot b \in S &\implies F \notin S; \quad n > F \implies F - n < 0 \implies F - n \notin S \implies n \in S \\ \implies \mathcal{G} \subset \{0, \dots, F\}, &\quad n \mapsto F - n \text{ carries } \mathcal{G} \text{ onto } S \cap \{0, \dots, F\} \\ \implies 2|\mathcal{G}| = F + 1 = (a-1)(b-1) &\implies |\mathcal{G}| = \frac{(a-1)(b-1)}{2}. \end{aligned}$$

□

A.3. Turning of a piecewise-smooth curve.

Definition A.3.1 (Turning). Let $c: [0, 1] \rightarrow \mathbb{R}^2$ be immersed and C^1 on each $[t_{k-1}, t_k]$, $0 = t_0 < \dots < t_m = 1$. On each piece the Gauss map $\dot{c}/|\dot{c}|: [t_{k-1}, t_k] \rightarrow S^1$ lifts to a continuous θ_k , unique up to $2\pi\mathbb{Z}$, and at an interior break the exterior angle $\varepsilon_k \in (-\pi, \pi)$ is the signed angle from $\dot{c}(t_k^-)$ to $\dot{c}(t_k^+)$. Then

$$\vartheta(c) = \sum_{k=1}^m (\theta_k(t_k) - \theta_k(t_{k-1})) + \sum_{k=1}^{m-1} \varepsilon_k \in \mathbb{R}.$$

Lemma A.3.2 (Additivity and invariance). *Turning is invariant under orientation-preserving reparametrisation, and for a concatenation with exterior angle ε at the join,*

$$\vartheta(c_1 * c_2) = \vartheta(c_1) + \vartheta(c_2) + \varepsilon, \quad \vartheta(\bar{c}) = -\vartheta(c).$$

Proof.

$$\begin{aligned} c \circ \varphi, \quad \dot{\varphi} > 0 &\implies \frac{(c \circ \varphi)'}{|(c \circ \varphi)'|} = \frac{\dot{c}}{|\dot{c}|} \circ \varphi \implies \theta \circ \varphi \text{ is a lift} \implies \vartheta(c \circ \varphi) = \vartheta(c) \\ \xrightarrow{A.3.1} \text{the breaks of } c_1 * c_2 &\text{ are those of } c_1, \text{ of } c_2, \text{ and the join} \implies \vartheta(c_1 * c_2) = \vartheta(c_1) + \vartheta(c_2) + \varepsilon \\ \dot{\bar{c}} = -\dot{c} \circ (1 - \cdot) &\implies \bar{\theta} = \theta \circ (1 - \cdot) + \pi \implies \vartheta(\bar{c}) = -\vartheta(c). \end{aligned}$$

□

Theorem A.3.3 (Umlaufsatz, Hopf). *A piecewise- C^1 embedded closed curve in $\mathbb{R}^2$ has $\vartheta = \pm 2\pi$, the sign being that of its orientation.*

APPENDIX B. REPRODUCIBILITY

B.1. The semigroup.

LISTING 1. `semigroup.py`: $d, S, \mathcal{G}$ and $t^g \Delta_{T(a,b)}$, with the checks of Theorem A.2.2, Proposition 5.3.2 and Theorem 6.3.3 at $(a,b) = (4,5)$.

```

1  """semigroup.py -- the arithmetic of S = <a,b>.
2
3  No plotting. The denominator, the semigroup and its gaps, and
4  Delta_{T(a,b)} from the generating identity.
5
6  python3 semigroup.py
7  """
8
9  def denominator(n, a, b):
10     return sum(1 for j in range(n // b + 1) if (n - b * j) % a == 0)
11
12  def semigroup(a, b):
13     F = a * b - a - b
14     S = [n for n in range(F + 2) if denominator(n, a, b)]
15     G = [n for n in range(F + 1) if not denominator(n, a, b)]
16     return S, G, F
17
18  def alexander(a, b):
19     """t^g Delta_{T(a,b)} as a coefficient list."""
20     g = (a - 1) * (b - 1) // 2
21     N = 2 * g + 1
22     d = [denominator(n, a, b) for n in range(N + a * b + 1)]
23     c = [0] * N
24     for n in range(N):
25         c[n] = d[n] - (d[n - 1] if n else 0) \
26             - (d[n - a * b] if n >= a * b else 0) \
27             + (d[n - a * b - 1] if n >= a * b + 1 else 0)
28     return c, g
29
30  if __name__ == "__main__":
31     a, b = 4, 5
32     S, G, F = semigroup(a, b)
33     c, g = alexander(a, b)
34     print("S =", S[:9], "...")
35     print("G =", G, " |G| =", len(G), " g =", g, " F =", F)
36     print("t^g D =", c)
37     print("support(D) =", sorted(n - g for n, x in enumerate(c) if x))
38     assert len(G) == g == (a - 1) * (b - 1) // 2
39     assert max(G) == F == 2 * g - 1
40     assert all((n in G) == (F - n in S) for n in range(F + 1))
41     assert all(x in (-1, 0, 1) for x in c)
42     assert c == c[::-1]
43     N = a + b
44     L = [[(i == j) - (i - j == a) - (i - j == b) + (i - j == a + b)
45           for j in range(N)] for i in range(N)]
46     d = [denominator(n, a, b) for n in range(N)]
47     Ld = [sum(L[i][j] * d[j] for j in range(N)) for i in range(N)]
48     print("L . d =", Ld)
49     assert Ld == [1] + [0] * (N - 1)
50     print("all checks pass")

```

B.2. The bigraded complex and the staircase.

LISTING 2. `staircase.py`: $(A(x_k), M(x_k))$ by the recursion of [10], and the E/S word of $\mathbb{L}_\perp$. Returns the A of Figure 14, Figure 15 and Example 5.2.1.

```

1  """staircase.py -- the numbers behind the left panel of the lattice figure.
2
3  No plotting. Two objects, both determined by S = <a,b>:
4
5  * the bigraded complex, i.e. the pairs (A(x_k), M(x_k)). With
6  d-hat = 0 the Maslov gradings are read off Delta_K by the recursion of
7  Ozsvath-Szabo, so (A, M) needs no diagram;
8  * the staircase word of S, one letter per n in 0..2g-1, E for n in S and
9  S for a gap.
10
11  python3 staircase.py

```

```

12 """
13
14 from semigroup import alexander, semigroup
15
16
17 def gradings(a, b):
18     """(A_k, M_k) for HFK-hat(T(a,b)), most positive Alexander first."""
19     c, g = alexander(a, b)
20     exps = [n - g for n in range(len(c) - 1, -1, -1) if c[n]]
21     assert all(c[e + g] == (-1) ** k for k, e in enumerate(exps)), \
22         "Delta must alternate"
23     M, out = 0, [(exps[0], 0)]
24     for k in range(1, len(exps)):
25         M = M - 2 * (exps[k - 1] - exps[k]) + 1 if k % 2 else M - 1
26         out.append((exps[k], M))
27     return out
28
29
30 def staircase(a, b):
31     """The E/S word of S = <a,b> on 0..2g-1."""
32     S, G, F = semigroup(a, b)
33     return "".join("E" if n in S else "S" for n in range(F + 1))
34
35
36 if __name__ == "__main__":
37     for a, b in ((2, 3), (3, 4), (4, 5)):
38         g = (a - 1) * (b - 1) // 2
39         AM = gradings(a, b)
40         print("T(%d,%d) g = %d" % (a, b, g))
41         print("  A =", [A for A, _ in AM])
42         print("  M =", [M for _, M in AM])
43         print("  staircase =", staircase(a, b))
44         # the worked tuple [7,1,2,2] = T(4,5)
45         AM = gradings(4, 5)
46         assert sorted(A for A, _ in AM) == [-6, -5, -2, 0, 2, 5, 6]
47         assert [M for _, M in AM][0] == 0
48         assert len(AM) == 7 # rk HFK-hat = p = 7
49         assert staircase(4, 5) == "ESSSEESSEES"
50         print("all checks pass")

```

B.3. The partition function and the tent.

LISTING 3. `partition.py`: the denominator against the line it samples, the tent of Lemma 6.3.1, and both transports of Theorem 6.3.3; the zeros of the denominator are the gaps.

```

1  """partition.py -- the numbers behind the partition figure and the
2  right panel of the lattice figure.
3
4  No plotting. The denominator against the line it samples, the periodic
5  bounded error, the box spline at the integers, and the zero set.
6
7  python3 partition.py
8  """
9
10 from fractions import Fraction
11 from semigroup import denominator, semigroup
12
13
14 def sampled(a, b, upto=None):
15     """rows (n, d(n), T(n), d(n) - T(n)) with T(u) = u_+/ab."""
16     upto = upto if upto is not None else 2 * a * b
17     return [(n, denominator(n, a, b), Fraction(n, a * b),
18             denominator(n, a, b) - Fraction(n, a * b))
19             for n in range(upto + 1)]
20
21
22 def tent(a, b, n):
23     """ab * B_{(a,b)}(n) by the closed form of the tent lemma."""
24     lo, hi = min(a, b), max(a, b)
25     return max(0, min(n, lo, lo + hi - n))
26
27
28 def nabla(f, a, b, n):
29     """(nabla_a nabla_b f)(n) for f defined on the integers."""
30     return f(n) - f(n - a) - f(n - b) + f(n - a - b)
31

```

```

32
33 if __name__ == "__main__":
34     a, b = 4, 5
35     S, G, F = semigroup(a, b)
36     rows = sampled(a, b)
37     print("n : d(n)  n/ab  d - n/ab")
38     for n, d, T, e in rows[:12]:
39         print("%2d : %d    %-6s %s" % (n, d, T, e))
40     print("zeros of d :", [n for n, d, _, _ in rows if d == 0], " = G :", G)
41
42     assert [n for n, d, _, _ in rows if d == 0] == G
43     assert max(G) == a * b - a - b
44     # d - T is ab-periodic and bounded by 1
45     e = {n: v for n, _, _, v in rows}
46     assert all(e[n] == e[n + a * b] for n in range(a * b))
47     assert all(abs(v) <= 1 for v in e.values())
48     # nabla_a nabla_b kills d off 0, and carries the truncated power to the tent
49     D = lambda n: denominator(n, a, b) if n >= 0 else 0
50     T = lambda u: Fraction(max(u, 0), a * b)
51     assert all(nabla(D, a, b, n) == (n == 0) for n in range(0, 40))
52     assert all(nabla(T, a, b, n) * a * b == tent(a, b, n) for n in range(0, 20))
53     print("tent  ab*B(n) =", [tent(a, b, n) for n in range(11)])
54     print("all checks pass")

```

B.4. The difference word.

LISTING 4. `word.py`: the difference word of the two closed-form branches, with Lemma 5.4.2 and both conjectures checked for $p = 3, \dots, 17$.

```

1  """word.py -- the numbers behind the difference-word figure.
2
3  No plotting.  The difference word of a tuple and the two conjectures it
4  is evidence for.  The Alexander vector A itself comes from the diagram
5  (cover.py); for the two closed-form branches of the two
6  corollaries of the closed-form families it is given here in closed form, so the checks run standalone.
7
8  python3 word.py
9  """
10
11 from staircase import gradings
12
13
14 def famA(p):
15     """[p,1,(p-3)/2,2]: K = T(v,v+1), v = (p+1)/2."""
16     v, g = (p + 1) // 2, (p * p - 1) // 8
17     A = [(p - 1) * (p + 1) // 8]
18     A += [(p - 4 * k + 5) * (p + 3) // 8 for k in range(2, v + 1)]
19     A += [(4 * k - 3 * p - 5) * (p + 1) // 8 for k in range(v + 1, p + 1)]
20     return A, v, (v, v + 1), g
21
22
23 def famB(p):
24     """[p,1,p-2,1]: K = T(2,p)."""
25     return [(p + 1 - 2 * k) // 2 for k in range(1, p + 1)], \
26            p - 1, (2, p), (p - 1) // 2
27
28
29 def word(A):
30     """d_k = A(x_k) - A(x_{k+1}), indices in Z/p."""
31     p = len(A)
32     return [A[k] - A[(k + 1) % p] for k in range(p)]
33
34
35 def three_letters(d):
36     """The c of the three-letter conjecture, or None."""
37     L = set(d)
38     for c in range(2, max(abs(x) for x in d) + 2):
39         if L <= {1, -c, c + 1}:
40             return c
41     return None
42
43
44 def three_gaps(d):
45     """Cyclic gap sets of each letter, the three-gap conjecture."""
46     p, out = len(d), {}
47     for u in sorted(set(d)):

```

```

48     P = [k for k in range(p) if d[k] == u]
49     out[u] = sorted({P[(i + 1) % len(P)] - P[i]} % p or p
50                     for i in range(len(P)))
51     return out
52
53
54 def ok_three_gaps(gaps):
55     for gs in gaps.values():
56         if len(gs) > 3 or (len(gs) == 3 and gs[0] + gs[1] != gs[2]):
57             return False
58     return True
59
60
61 if __name__ == "__main__":
62     for name, fam, lo in (("A", famA, 5), ("B", famB, 3)):
63         print("family %s" % name)
64         for p in range(lo, 18, 2):
65             A, v, (a, b), g = fam(p)
66             d = word(A)
67             c = three_letters(d)
68             gaps = three_gaps(d)
69             assert len(A) == p and max(A) == g
70             assert sorted(A) == sorted(x for x, _ in gradings(a, b)), \
71                 "A is not the support of Delta_{T(%d,%d)}" % (a, b)
72             assert sum(d) == 0
73             assert d[0] == d[v - 1] == 1          # the rainbow steps
74             assert c is not None and ok_three_gaps(gaps)
75             print("  p=%-3d T(%d,%d)  g=%-3d v=%-2d  letters {1,%d,%d}  "
76                   "gaps %s" % (p, a, b, g, v, -c, c + 1,
77                               {u: gs for u, gs in gaps.items()}))
78     print("all checks pass")

```

B.5. Changing basis on the torus.

LISTING 5. develop.py: the change of basis of Example A.1.7, unimodular and gcd-preserving, carrying $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ to $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

```

1  """develop.py -- the change of basis behind the development figures.
2
3  No plotting.  H rewrites the square's coordinates in the meridian-longitude
4  basis, H(u,v) = (u, v-u).  It is unimodular, hence an automorphism of
5  H_1(T^2) = Z^2, and it preserves primitivity, so the class of beta in the
6  new basis is the class of the knot on the Heegaard torus.
7
8  python3 develop.py
9  """
10
11 from math import gcd
12
13 H = ((1, 0), (-1, 1))          # (u, v) -> (u, v - u)
14 HINV = ((1, 0), (1, 1))
15 I2 = ((1, 0), (0, 1))
16
17 def det2(M):
18     return M[0][0] * M[1][1] - M[0][1] * M[1][0]
19
20
21 def mul(M, N):
22     return tuple(tuple(sum(M[i][k] * N[k][j] for k in range(2))
23                         for j in range(2)) for i in range(2))
24
25
26 def act(M, c):
27     return tuple(sum(M[i][k] * c[k] for k in range(2)) for i in range(2))
28
29
30 def torus_class(c):
31     """|a|, |b| of the curve of class c = (a, b); it is a knot iff coprime."""
32     a, b = abs(c[0]), abs(c[1])
33     return (a, b) if gcd(a, b) == 1 else None
34
35
36 if __name__ == "__main__":
37     assert det2(H) == 1 and mul(H, HINV) == I2
38     # unimodular, so primitivity is preserved in both directions

```



```

40 for u in range(-6, 7):
41     for v in range(-6, 7):
42         if (u, v) != (0, 0):
43             assert gcd(*act(H, (u, v))) == gcd(u, v)
44 # D[3,1,1,1]: the development reports surface class (2,-3)
45 square = act(HINV, (2, -3))
46 print("surface class (2,-3) <- square class", square)
47 print("knot type T%s" % (torus_class((2, -3)),))
48 assert square == (2, -1)
49 assert torus_class((2, -3)) == (2, 3)
50 print("all checks pass")

```

B.6. The diagram on the torus.

LISTING 6. `torus.py`: the surface, the camera and the arcs of Figure 13, with the three checks that precede emission.

```

1 import math
2
3 R, r, P = 1.00, 0.42, 6
4 STEP = 2 * math.pi / P
5 F1 = math.pi / 2 - STEP # the two bigons straddle f = pi/2
6 FS = [F1 + h * STEP for h in range(P)]
7 DL = 0.58 # how far a rainbow bulges off alpha
8
9 def surface(t, f, e):
10     """S(t,f) seen from d = (0,-cos e, sin e), screen up (0, sin e, cos e)."""
11     ce, se = math.cos(e), math.sin(e)
12     x = (R + r * math.cos(f)) * math.cos(t)
13     y = (R + r * math.cos(f)) * math.sin(t)
14     return x, y * se + r * math.sin(f) * ce
15
16 def front(t, f, e):
17     """n . d for the outward normal n; the silhouette is front = 0."""
18     ce, se = math.cos(e), math.sin(e)
19     return -math.cos(f) * math.sin(t) * ce + math.sin(f) * se
20
21 def rainbow(fa, fb, base, n=200):
22     """An arc leaving a wall at fa, bulging DL into the annulus, back at fb."""
23     s = 1 if base == 0 else -1
24     return [(base + s * DL * math.sin(math.pi * i / n),
25             fa + (fb - fa) * (1 - math.cos(math.pi * i / n)) / 2)
26             for i in range(n + 1)]
27
28 def band(fa, n=300):
29     """L_h to R_{h+1}: once round the longitude, rising one step in f."""
30     return [(2 * math.pi * i / n, fa + STEP * i / n) for i in range(n + 1)]
31
32 BETA = [rainbow(FS[0], FS[1], 0.0), rainbow(FS[1], FS[2], 2 * math.pi)] \
33         + [band(FS[j]) for j in range(2, P)]
34
35 if __name__ == "__main__":
36     e = math.radians(52.0)
37     # beta is one closed curve: the gluing L_h ~ R_h induces a P-cycle
38     nxt = {1: 2, 2: 3}
39     nxt.update({j: j % P + 1 for j in range(3, P + 1)})
40     orbit, h = [1], 1
41     while nxt[h] not in orbit:
42         h = nxt[h]; orbit.append(h)
43     assert sorted(orbit) == list(range(1, P + 1)), orbit
44     # beta meets alpha in exactly the P generators
45     hits = {round(f % (2 * math.pi), 9) for arc in BETA for t, f in arc
46             if min(t, abs(2 * math.pi - t)) < 1e-9}
47     assert len(hits) == P
48     # both bigons face the viewer entirely
49     lo = min(front(t, f, e) for arc in BETA[2:] for t, f in arc)
50     print("P = %d, elevation 52, min frontness over the bigons %.3f" % (P, lo))
51     assert lo > 0.12
52     print("all checks pass")

```

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